

Structural transformation and female labour participation in agriculture: Evidence from Mexico, 1900–2024

Abstract

This paper offers an in-depth analysis of female labour participation in Mexico's agricultural sector. Using probit regressions and microdata from labour force surveys between 2005 and 2019, it finds that women report a low likelihood of labour force participation in Mexico's agricultural regions. Notably, these results do not stem from any underestimation of women's agricultural work. This prompts a critical question: why is female labour participation so low in Mexican agriculture? Using data from the 2022 Mexican Agricultural Census and the Central Bank of Mexico, the research explores three potential mechanisms associated with low female labour participation in agriculture. First, a demand-side mechanism: the use of machinery in Mexican agriculture is reducing labour demand for female workers. Second, a supply-side mechanism: the influx of remittances from Mexicans in the USA is lowering women's labour supply in rural areas. Third, the draught-animal mechanism: if agricultural activities employ animal traction, they become male-dominated because of the upper-body strength required to control the animals. However, the exploratory analysis shows that the low female labour participation in Mexican agriculture did not start recently as a consequence of any of these mechanisms. Instead, the paper documents Mexico's structural transformation from 1900 to 2024 and finds that the share of women in the agricultural workforce has been consistently low since the early twentieth century, when Mexico was still a predominantly agrarian economy. Therefore, the paper concludes that low female labour participation in Mexican agriculture is not a recent development but, rather, a persistent historical pattern.

1.1 Introduction

A standard pattern of economic development across countries is the structural transformation of their labour markets. As the economy evolves, workers move from agriculture to industrial and service activities with higher incomes and productivity rates (Herrendorf et al., 2014; McMillan et al., 2014). Lewis (1954) argued that the labour reallocation of men from subsistence farming to industrial jobs would not cause labour shortages in agriculture, as wives and daughters could act as a reserve workforce to replace them. Later on, Cernea (1978) was the first to document the feminisation of agriculture in Romania by showing how structural change, rapid industrialisation and rural–urban migration led to women’s predominance in agricultural work relative to men. Recently, Slavchevska et al. (2016) and Bryceson (2019) have identified the feminisation of agriculture in more countries, as women comprise over 50% of their national agricultural workforce¹. Contemporary research also argues that the outmigration of men from agriculture to non-agricultural sectors increases the labour supply and the concentration of women in agricultural activities (Baada & Najjar, 2020; Kawarazuka et al., 2020; Mu & van de Walle, 2011; Mukhamedova & Wegerich, 2018; Radel et al., 2012; Slavchevska et al., 2020), and that structural transformation may be leading to a feminisation of agriculture (Slavchevska et al., 2019).

Some researchers also indicate that over the past few decades, there has been a trend towards the feminisation of agricultural activities in Latin American countries, since women are increasingly performing traditionally male tasks such as land preparation and cash crop production (Lastarria-Cornhiel, 2006), while they also display particularly high participation in agricultural enterprises exporting vegetables, fruits and flowers (Deere, 2005). Data from the World Bank’s World Integrated Trade Solution indicate that, in 2023, Mexico was the top exporter of food products to the USA (WITS, 2023a), while ranking seventh in global fruit exports (WITS, 2023b), and 10th in global vegetable exports (WITS, 2023c). These World Bank statistics also indicate that Mexico is by far the largest exporter of fruits and vegetables in Latin America, surpassing other leaders in agricultural exports such as Brazil and Argentina. Therefore, this research examines the feminisation of agriculture using Mexico as a case study, since this country has followed the standard pattern of structural transformation—a reallocation of labour from agriculture to industry and services—while also having the most export-oriented agricultural sector in the region.

¹ These countries are Armenia, Azerbaijan, Bhutan, Gambia, Georgia, Ghana, Guinea, Kenya, Malawi, Mali, Nepal, Rwanda, Tajikistan, Tanzania, Uganda, Vietnam, Zambia and Zimbabwe.

Recently, Doss & Gottlieb (2025) examined gendered patterns of structural transformation in agricultural labour and found considerable heterogeneity across middle-income economies, suggesting that each country follows its own story of how agricultural work is divided between men and women. For instance, in India, the evidence indicates that female labour participation rates (FLPRs) have been declining thanks to women's exit from agriculture (Chand & Srivastava, 2014), combined with a lack of suitable job opportunities in industry and services (Chatterjee et al., 2015). Specifically, Deshpande & Singh (2024) argue that structural transformation in India occurred alongside insufficient job creation, resulting in a period of 'jobless growth' that failed to absorb women into productive non-farm employment. In contrast, Roncolato (2016) finds that, in South Africa, female labour participation reaches its highest levels in the municipalities where agricultural jobs predominate. This finding is relevant because, in South Africa, women do not account for more than 50% of the agricultural workforce (see Figure 1.2 in Appendix A, p. 42). This suggests that it is not enough to examine the gender composition of the agricultural workforce; it is also necessary to study female labour participation in local labour market structures where agricultural jobs predominate.

This discussion raises a question: Is female labour participation relatively high in Mexico's agricultural regions? To answer it, this study follows the methodology of Roncolato (2016), by employing probit regressions to estimate the probability of female labour participation as a function of the share of non-agricultural jobs at the municipal level. This model specification is grounded in the structural transformation literature, since the reallocation of labour out of agriculture is typically associated with a rise in GDP per capita (Herrendorf et al., 2014). To follow this empirical strategy, the analysis relies on repeated cross-sectional microdata from Mexico's labour force survey for the first quarters of 2005, 2010, 2015, and 2019.

The regression results do not show the same trend that Roncolato (2016) found in South Africa. In Mexico, the probability that women participate in the labour force decreases considerably if they live in agricultural regions of the country. Official statistics also report that, in 2019, roughly 90% of the Mexican agricultural workforce were men and only 10% women (see Figure 1.5 in Appendix B, p. 52). Further, among working women in Mexico, 80% are employed in services, 17% in industry, and only 3% in agriculture (see Figure 1.6 in Appendix B, p. 52).

Notably, this paper argues that considerable underestimations of women's work in agriculture are unlikely in Mexico's labour force surveys.² In Mexico, a person is classified as part of the workforce if they report doing subsistence farming, agriculture for self-consumption, unpaid farm work, working at least 1 hour per week, or actively seeking employment. Despite these inclusive criteria, very few Mexican women report working in agriculture.

These findings prompt a second question: why is female labour participation in agriculture so low in Mexico? The study explores three potential mechanisms that might explain this pattern. First, a supply-side mechanism: Mexican women in rural areas are less willing to supply labour due to income effects from remittance inflows (Amuedo-Dorantes & Pozo, 2006). Second, a demand-side mechanism: agricultural machinery may be generating a reduction in the labour demand for female workers in agriculture (Afridi et al., 2023; Takeshima, 2024). Third, the draught-animal mechanism: if agricultural activities use animal traction, they become male-dominated because of the upper-body strength required to control the animals.

To explore the validity of these mechanisms, this paper uses data from Mexico's Central Bank to capture remittance inflows at the municipal level, and it uses data from the 2022 Mexican Agricultural Census to estimate the use of agricultural machinery (tractors), the use of draught animals in agrarian activities at the municipal level, and the share of women in the agricultural workforce at the municipal level. However, the exploratory analysis does not find evidence to conclude that low female labour participation in Mexican agriculture is a recent phenomenon arising from remittance inflows (see Figure 1.10, p. 59), tractor use (see Figure 1.11, p. 60), or draught animal use (see Figure 1.12, p. 60).

As the lack of female labour participation in Mexican agriculture does not stem from any of these mechanisms, the study explores decennial census data from 1900 onwards to analyse Mexico's structural transformation process and women's participation in agriculture since the beginning of the twentieth century. The data indicate that, in 1900, Mexico was an agrarian country and the agricultural sector accounted for 60%–70% of the total workforce for several decades thereafter. It was not until 1990 that the agricultural sector was surpassed by industry and services (see Figure 1.14 in Appendix B, p. 62). Moreover, census data show that women have represented only a small proportion of Mexico's agricultural labour force since 1900 (see Figure 1.13 in

² The ENOE questionnaire has a series of questions to determine whether an individual is genuinely not part of the labour force: 1) "Did you work for at least one hour last week?"; 2) "Regardless of what you just told me, did you spend at least one hour last week on an activity that provided you with income?"; 3) "Regardless of what you just told me, did you spend at least one hour last week helping with the land or business of a family member or another person?"; 4) "Although you already told me that you did not work last week, do you have any job, business, or engage in any activity on your own account?". Only those answering negatively to all these questions are considered non-working individuals.

Appendix B, p. 61). Hence, the research concludes that low female participation in Mexican agriculture is a historical pattern, rather than a recent phenomenon.

The research presents valuable evidence for the literature on female labour participation in agriculture. At the cross-country level, it examines the agricultural feminisation theory and presents data showing that very few countries have a predominantly female agricultural sector (see Figure 1.2 in Appendix A, p. 42). In fact, the general trend indicates that a lower percentage of agricultural jobs at the national level is associated with a lower proportion of women in the agricultural workforce. At the within-country level, the study examines Mexico's structural transformation and finds that this process was not followed by any agricultural feminisation either. Finally, the study explores various mechanisms often cited as potential explanations for low female labour participation in agriculture and finds that the pattern observed in Mexico is not rooted in any of these contemporary factors. Taken together, these findings reveal that agricultural feminisation is not a pattern of the structural transformation process, and that studying women's involvement in agriculture requires examining both contemporary and historical data sources.

The following sections offer further insights into women's participation in agricultural labour markets. The final goal is to discuss theories and present empirical evidence to deepen our understanding of this research topic. The remainder of the paper is organised as follows: Section 2 presents a review of the literature; Section 3 details the methodology; Section 4 presents the results; Section 5 presents a discussion; and Section 6 offers concluding remarks.

1.2 Literature review

This section reviews different strands of literature on female labour participation in agricultural activities. The first part clarifies the definition and the conceptual debates surrounding the agricultural feminisation hypothesis. The second discusses and contrasts evidence regarding high and low levels of female involvement in agricultural activities across countries. The third part explores competing mechanisms that may explain the lack of female participation in agriculture. The fourth focuses on studies indicating that some countries have historically low female labour participation in agriculture and discusses the different potential reasons behind this pattern.

1.2.1 The feminisation of agriculture

The reallocation of labour from agriculture to non-agricultural activities is a common example of structural transformation across countries. In the model of economic development with unlimited supplies of labour, Lewis (1954) argued that the shift of male labour from subsistence farming to industry would not create labour shortages in agriculture, as wives and daughters who were not in the labour force could take over their work in this sector.³ Boserup (1970) further predicted that rapid population growth and limited capital in developing countries would restrict the use of capital-intensive technologies, and this would increase the labour demand for female workers in agriculture. A few years later, Cernea (1978) documented the ‘feminisation of agriculture’ in his study of structural change in Romania. He argued that rapid industrial growth and rural–urban migration had led to a predominance of women in agriculture. Using data from 1974, he showed that women made up 63% of the labour force in Romania’s agricultural cooperatives.

Unfortunately, the term ‘feminisation of agriculture’ has been used inconsistently, as scholars have assigned divergent meanings to it. Some of them use it to describe cases in which women represent more than 50% of the agricultural labour force (Doss & Gottlieb, 2025; Najjar et al., 2018). Others employ it more loosely to indicate the rising participation of women in agriculture, even if they are still a minority of the labour force in this sector (Abdelali-Martini et al., 2003; Agarwal, 2011). To address this ambiguity, Slavchevska et al. (2016) distinguish between feminised agriculture (where women constitute the majority of the agricultural workforce) and an ongoing feminisation of agriculture (where the proportion of women in agriculture is rising significantly). Nevertheless, this ambiguity weakens analytical clarity. While the concept was initially coined to describe the predominance of women in agriculture, some studies now use it to describe the rising participation of women in agriculture.

The implications of the feminisation of agriculture are also contested. Some scholars frame it in optimistic terms, arguing that expanded roles on family farms may enhance women’s voice within households, communities, and agricultural governance (Anderson et al., 2021; Krumbiegel et al., 2020). Others emphasise that women are often integrated into agriculture as seasonal or casual labourers, typically occupying the lowest-paid, most insecure, and least protected positions within the rural labour markets (Fabry et al., 2022; Fabry et al., 2024). Therefore, female labour participation in agriculture should not be assumed to be inherently positive. It depends both on

³ Lewis (1954) argued that a country could initially experience an enormous expansion of new industrial jobs without any shortage of unskilled labour becoming apparent in the labour market. His argument rests on the premise that sources of unlimited labour supply include women in the household, petty traders, and retainers (domestic servants and commercial assistants).

the type and quality of work that women perform and on the framework used to assess whether such participation is beneficial or detrimental to them.

Researchers have studied the feminisation of agriculture across countries and regions. Asadullah & Kambhampati (2021) state that, while female labour force participation remains low in developing countries, women's involvement in agriculture is growing globally. Furthermore, Slavchevska et al. (2016) present a cross-country analysis of the feminisation of agriculture, which finds that women make up over 50% of agricultural workers in Azerbaijan, Armenia, Bhutan, Georgia, Tajikistan, Nepal, Vietnam, The Gambia, Guinea, Uganda, and Zimbabwe. They conclude that male outmigration is one of the main drivers for the feminisation of agriculture. Similarly, contemporary research has identified that the outmigration of men from agriculture increases the concentration of women in agricultural activities (Baada & Najjar, 2020; Kwarazuka et al., 2020; Mukhamedova & Wegerich, 2018; Radel et al., 2012; Slavchevska et al., 2020).

Other studies analysing the feminisation of agriculture have focused either on a single country or on several countries within a specific region. In Syria, Abdelali-Martini et al. (2003) document that the absence of male workers as a result of rising migration rates has made the agricultural workforce predominantly female. In Egypt, Binzel & Assaad (2011) studied the labour supply responses of wives left behind by Egyptian men working abroad. Their results show that rural women with migrant partners are more likely to engage in unpaid family work and subsistence agricultural activities than are women with non-migrant partners.

In India, Pattnaik et al. (2018) report a feminisation of agriculture primarily driven by male outmigration from rural areas. They find that women's increased participation in agriculture correlates with indicators of poverty rather than signs of empowerment. Hence, they conclude that this Indian trend should be reconceptualised as a feminisation of agrarian distress. Using Nepal as a case study, Gartaula et al. (2010) argue that the feminisation of agriculture resulting from male out-migration requires a distinction between labour feminisation and managerial feminisation, while Bhawana & Race (2020) argue that the feminisation of agriculture in Nepal is being constrained by traditional gender norms. In Bangladesh, de Brauw et al. (2021) investigate the out-migration of household members and concluded that in households experiencing shortages in labour supply, there is a greater use of female household workers, but a reduced share of female hired workers, while there is no evidence of enhanced women's empowerment in these households. Meanwhile, Bacud et al. (2021) show that male outmigration in Vietnam is increasing women's participation in agriculture, but as unpaid workers, not as salaried workers.

In Southeast Asia, Akter et al. (2017) conducted focus group discussions and concluded that women play an active role in agricultural groups in Thailand and the Philippines, but not in Indonesia and Myanmar. In China, Mu & van de Walle (2011) find that the transition out of agriculture has increased the labour supply of left-behind women living in rural areas, as these women report substantially more hours worked in agriculture. Similarly, de Brauw et al. (2013) highlights a feminisation of Chinese agriculture since the late 1990s, noting that women have increased their labour supply on farms and their involvement in agricultural management.

In Sub-Saharan Africa, Bryceson (2019) studied gender patterns in the ‘deagrarianisation’ of eight African countries (Ghana, Zambia, Mali, Tanzania, Rwanda, Kenya, Malawi, and Uganda) between 1980 and 2015. The author notes that, during this period, men progressively withdrew from agricultural activities, particularly cash cropping, shifting towards non-agrarian activities—mostly in local services, with a few in artisanal mining—to seek new income-generating opportunities. Three decades later, this transition led to a feminisation of agriculture, with women dominating that sector while men specialised in emerging non-agricultural sectors. More specifically, in six of the eight countries studied (Zambia, Tanzania, Rwanda, Kenya, Malawi, and Uganda), women constitute the primary workforce in agriculture, whereas men predominate in non-agricultural sectors. Meanwhile, Ghana reports an almost gender-balanced agricultural sector. In contrast, Mali, a Muslim country, is the only one still showing strong male dominance in agriculture; this may be thanks to female seclusion practices in Islamic traditions. In addition, Ali et al. (2016) as well as Abdisa et al. (2024) argue that women constitute 50% of the agricultural workforce in Sub-Saharan Africa. Meanwhile, Palacios-Lopez et al. (2017) estimate that the proportion of women in crop production exceeds 50% in Malawi, Tanzania and Uganda, but they constitute a smaller proportion in Nigeria (37%), Ethiopia (29%) and Niger (24%). Finally, Haug et al. (2021) examine the feminisation of agriculture in six African countries (Malawi, Rwanda, South Africa, Kenya, Tanzania, Ethiopia) and find that women are experiencing increased workloads without corresponding improvements in their well-being.

In Latin America, Deere (2005) notes a feminisation of agriculture in recent decades, driven by the rising participation of rural women as own-account and wage workers. She argues that the rise is concentrated in export-oriented businesses producing fresh vegetables, fruits and flowers. Furthermore, Lastarria-Cornhiel (2006) states that the traditional gender division of labour in Latin American agriculture is weakening, as women are increasingly taking on tasks such as land preparation and cash-crop production. Moreover, Radel et al. (2012) conducted field research in

some Mexican southeastern regions and concluded that there is a feminisation of agriculture associated with male Mexican migration to the USA.

On the other hand, Kawarazuka et al. (2022) challenge the argument that the feminisation of agriculture constitutes a global trend, dismissing it as a myth that needs to be debunked. Doss & Gottlieb (2025) also dispute the notion of a widespread feminisation of agriculture, asserting that cross-country data do not support the argument that women are becoming the dominant labour force in the agricultural sector. Their analysis of labour force surveys from 85 countries reveals that men remain the majority of agricultural workers in most cases. Nevertheless, Heckert et al. (2021) use sex-disaggregated data from 36 low- and middle-income countries and find that structural transformation (measured as the percentage of the GDP coming from non-agricultural activities) is positively associated with on-farm employment of young women and negatively associated with on-farm employment of young men. They argue that these results may indicate that young women are taking on men's roles in agriculture as they shift to non-agricultural jobs.

The objective of this research is not only to discuss theory but also to contrast it with the latest available data. Using World Bank Data, I have calculated the proportion of men in the agricultural workforce of more than 170 economies. Figure 1.2 (see Appendix A, p. 42) shows that most countries have an agricultural labour force predominantly composed of men, with only a few exceptions. This is an important contribution to this research topic, as the study by Doss & Gottlieb (2025) considered 85 countries, while the present study offers data from more than 170 economies. Additionally, this analysis reveals an interesting trend: the lower the percentage of agricultural jobs across countries, the lower the proportion of women participating in the agricultural sector. To preserve the focus of the literature review, all the details about this figure are discussed in Appendix A. However, the data suggest that, in most countries, the majority of agricultural jobs are performed by men.

1.2.2 Female labour participation in agricultural labour markets

Numerous cross-country studies show that FLPRs are at their peak in low-income countries where agricultural jobs predominate, decline in middle-income countries that are typically experiencing an expansion of industrial activities, and rise again in high-income countries with a service-oriented economy (Goldin, 1994; Heath & Jayachandran, 2016; Jayachandran, 2020; Psacharopoulos & Tzannatos, 1989; Tam, 2011; Uberti & Douarin, 2023; Verick, 2014; Verme, 2015).

However, most cross-country studies compare FLPRs with GDP per capita rather than sectoral employment composition. Figure 1.3 (p. 46) confirms that across countries, there is a positive relationship between FLPRs and a higher share of agricultural jobs, a negative relationship between FLPRs and a higher share of industrial jobs, and a positive relationship between FLPRs and a higher share of service jobs. Notably, countries with the highest percentages of agricultural jobs consistently report the highest FLPRs worldwide. This is an overlooked trend, as the literature fails to note that the rise of FLPRs in high-income countries with service-oriented economies remains below the levels observed in low-income countries with agricultural labour markets.

Goldin (1994) argued that FLPRs tend to be high in low-income economies where agricultural jobs predominate. However, this argument does not suggest that women will eventually constitute the majority of the agricultural workforce. It simply argues that, in poor economies where agricultural jobs predominate, women tend to report high FLPRs. Nevertheless, this theory does not attribute this pattern exclusively to labour market structures characterised by agricultural predominance. Other factors also push women towards agricultural work as a subsistence strategy in low-income settings. The theory indicates that low household incomes, high fertility rates and dependency ratios, the compatibility of agricultural work with childcare responsibilities, and the predominance of crops that traditionally rely on female labour (e.g. sugarcane, rice, cotton and tree crops) all contribute to high female participation in agriculture.

Within-country evidence shows a pattern similar to that described by Goldin. Roncolato (2016) conducted a study in South Africa to estimate the probability of women participating in the workforce depending on the share of agricultural and non-agricultural jobs in the municipality where they were living. In other words, her study analysed the relationship between local labour market structures and female labour participation patterns. The results showed that South African women living in agricultural municipalities were the most likely to participate in the labour force. Therefore, her evidence suggests that female labour participation may be high not only in poor countries with large agricultural sectors but also in agricultural regions of middle-income countries.

Nevertheless, it seems that women's engagement in agricultural activities during the structural transformation process varies considerably across countries. Doss & Gottlieb (2025) highlight that middle-income economies exhibit divergent patterns in the gendered division of agricultural labour, with country-specific factors shaping the outcomes. This heterogeneity across developing countries indicates that there is no single story to explain female participation in agriculture during structural changes in the labour market and thus calls for disaggregated analyses that capture the particularities of each country.

Evidence from India can serve as the first example. Since the liberalisation reforms in 1991, India has undergone a structural transformation process accompanied by sustained economic growth, declining poverty rates, falling fertility and rapid increases in educational attainment, but also a significant decline in FLPRs during the same period (Deshpande & Singh, 2024). Research suggests this decline in FLPRs stems from two concurrent factors: women's exit from agriculture (Chand & Srivastava, 2014), and insufficient job creation in industry and services to absorb them in other sectors (Chatterjee et al., 2015). In line with these arguments, Mehrotra & Parida (2017) argue that, between 2004 and 2012, there was a decline of 37 million jobs in agriculture, of which 31 million were held by women. Their analysis further notes that nine million women in India transitioned into non-agricultural sectors, while the remainder withdrew entirely from the labour force, either to pursue education or to undertake domestic responsibilities.

Turning to the Chinese context, Zhang et al. (2004) explain that, since the initiation of economic reforms in the late 1970s, China has undergone both sustained rapid growth and profound structural transformation, with the expansion of non-agricultural employment representing a particularly salient feature of this transition. Their analysis focuses on studying the development of rural labour markets in China and their gender implications. While in 1981 only 16% of jobs in rural labour markets were non-agricultural, by 2000 nearly 50% of jobs were non-agricultural. In contrast to the Indian experience, their findings indicate that structural transformation in China has facilitated women's transition out of agriculture into non-farm jobs. While in 1990 only 10% of women in rural areas had non-agricultural jobs, by 2000 more than 30% of them had off-farm employment. Assuming that participation in non-agricultural jobs is a sign of increased welfare, the authors conclude that women in rural labour markets have benefited from China's structural transformation.

In Ethiopia, Bezu et al. (2012) conducted a comparative analysis of rural households, distinguishing between those engaged exclusively in agricultural activities and those with diversified incomes from both farm and non-farm employment. The authors contend that female labour force participation in Ethiopian agriculture is considerably constrained by prevailing cultural and gender norms. Consequently, they argue that in households relying solely on agriculture, adult female labour is typically underutilised or partially idle. In contrast, households that diversify their income with non-farm activities show higher returns on female labour participation. They argue that in these households, women participate in petty trade or wage employment, thereby enabling a more productive utilisation of female labour. The study argues that non-farm employment offers

particular benefits for women in Ethiopia, providing a productive outlet for labour that would otherwise be constrained by the gender roles established in the agricultural division of labour.

In Egypt, Suleiman (2025) examines the relationship between economic development, structural transformation and changes in female labour force participation between 1989 and 2022. The study notes that before 1990, agriculture constituted the principal source of employment in Egypt, followed by services, while the industrial sector accounted for the smallest share of jobs. Conversely, by 2022, the employment structure had undergone a sectoral recomposition: services accounted for over 50% of jobs, industry for more than 25%, while agriculture had been relegated to third place, representing less than 20% of the total workforce. Concurrent with this shift in the labour market structure, the author identifies a contraction in the proportion of women in the total workforce, from 28% in 1989 to 17% in 2022. Using provincial-level data, his research finds that the share of women in the total workforce is positively correlated with a higher percentage of agricultural jobs and negatively correlated with a higher percentage of industrial jobs⁴. In addition, the author indicates that a higher percentage of white-collar services is associated with greater female labour participation. Therefore, the author contends that the decline in female labour force participation in Egypt is linked to structural changes in the sectoral composition of employment. Specifically, the author argues that women's exit from agriculture has not been compensated by a comparable reallocation of female workers into other sectors.

In sum, South Africa exhibits high FLPRs in its agricultural municipalities. In Egypt, there is a positive relationship between the percentage of jobs in agriculture and the share of women in the total labour force at the provincial level. In rural Ethiopia, gender norms constrain female participation in agriculture, but households that diversify their income with non-farm activities may employ female labour in non-agricultural jobs. In China, the structural transformation process has facilitated women's transition from agriculture to non-agricultural jobs. Conversely, in India and Egypt, the decline in women's participation in agriculture has not been matched by comparable increases in their participation in industry or services. Instead, women are leaving the workforce to pursue education or engage in domestic chores. Hence, these country examples suggest that each developing country has a distinct history regarding its evolution in the gendered division of the agricultural workforce. Ultimately, female participation in agricultural labour markets varies across national contexts, as no single pattern emerges consistently from the literature on this topic.

⁴ The analysis presented by Suleiman (2025) shows a negative relationship between the share of industrial jobs at the provincial level and the share of women in the total labour force. Nevertheless, a subsequent analysis breaks down the industrial sector into three sub-sectors: i) manufacturing, ii) construction, and iii) mining and utilities. This second analysis shows a positive relationship between manufacturing jobs and FLFPS (female labour force participation share).

1.2.3 Mechanisms behind low female labour participation in agriculture

This paper studies women's participation in Mexico's agricultural labour markets. The analysis reveals considerably low levels of female engagement in this sector. Accordingly, this subsection reviews other studies that present empirical evidence identifying competing mechanisms that may explain the low labour participation of women in Mexico's agricultural sector.

1.2.3.1 Remittances

The first mechanism is grounded in a strand of literature examining the potential income effects of remittances on the labour supply decisions of men and women. In this context, a standard prediction in labour supply theory is that the income effect arises because remittances represent an exogenous increase in non-labour income that shifts the household's budget constraint, enabling members to consume more leisure and reduce labour supply while maintaining or increasing overall household consumption. This supply-side mechanism warrants examination, as Mexico is the second-largest recipient of remittances globally, just behind India. In 2022, Mexico received over US\$61 billion in remittances, while India received over US\$111 billion (International Organization of Migration, 2024). Nevertheless, Mexico's population of 132 million inhabitants is at least ten times smaller than India's 1,400 million, meaning that remittance inflows per capita are considerably higher in Mexico than in India.

Several studies of different countries have found that remittances are typically associated with a lower female labour supply, particularly among women living in rural areas (Acosta, 2006; Adams JR, 2011; Asiedu & Chimbar, 2020; Azizi, 2018; Jadotte & Ramos, 2016). In the case of Mexico, Amuedo-Dorantes & Pozo (2006) studied how male and female labour supplies respond to increases in remittance income. Their findings show that, while the labour supply of Mexican men remains largely unaffected, the labour supply of Mexican women declines significantly in response to remittance inflows, particularly if they live in rural areas.⁵ The authors suggest that this

⁵ The findings of my study seem to differ from those of Amuedo-Dorantes & Pozo (2006) who report that remittances are associated with a withdrawal from informal employment and unpaid work among Mexican women in rural areas. This apparent discrepancy can be reconciled by considering the sectoral composition of women's work. Amuedo-Dorantes and Pozo examine female labour force participation across all sectors (agriculture, industry, and services), whereas this research focuses on FLP in agriculture. Given that over 80% of Mexico's female labour force is employed in services, and that FLP in agriculture has remained historically low since 1900, the withdrawal from informal work they document likely occurs predominantly in services. In the coming sections, this research shows that Mexican municipalities do not show a negative relationship between remittances and female labour participation in agriculture. Instead, this research finds that women's limited engagement in Mexico's agricultural sector is not a recent development, but a persistent feature documented since 1900.

income effect, driven by remittances, allows Mexican women to withdraw from informal employment or unpaid work. The authors argue that this pattern resembles the neoclassical model of labour–leisure choice proposed by Killingsworth (1983), which posits that non-labour income, such as remittances, relaxes household budget constraints, increases reservation wages, and generates an income effect that reduces labour supply. Hence, this evidence suggests that remittances could be one reason for the low female labour participation in Mexican agriculture.

1.2.3.2 Agricultural machinery

The adoption of agricultural machinery represents a fundamental transformation of rural labour markets. Goldin (1994) briefly discussed that the mechanisation of agricultural production may reduce the labour demand for manual labour traditionally performed by women, such as weeding. Mehrotra & Parida (2017) also argued that agricultural mechanisation had contributed to the decline of women in the agricultural labour force in India, while Afridi et al. (2023) confirmed that agricultural mechanisation in India reduced the labour demand for precision-intensive manual tasks (e.g. weeding) in which Indian women were typically concentrated. Takeshima (2024) also finds that the adoption of tractors and harvesters has accelerated women's exit from agricultural jobs across seven African and Asian countries. These findings suggest a second mechanism explaining cross-country variation in female participation in agriculture. High participation in South Africa may reflect lower mechanisation, whereas low participation in Mexico may indicate more advanced mechanisation, reducing demand for female workers in labour-intensive roles.

1.2.3.3 Animals

The literature also identifies gendered patterns in livestock-related agricultural tasks. The draught-animal mechanism was proposed by Boserup (1970), who argued that, if land preparation for sowing is done using draught animals, this activity will predominantly be performed by men because of the upper-body strength required to control the animals. Consequently, this practice will reduce the need for weeding, a task typically performed by women. As a result, women's participation in agricultural labour declines on farms that employ draught animals; in some cases, they are entirely released from farm tasks.

Although the draught animal mechanism is the one formally explored in this research, it is worth highlighting other studies that have examined gendered roles in the management of animals

for agricultural purposes. For instance, Raney et al. (2011) argue that men predominantly engage in herding grazing animals. Meanwhile, Tangka et al. (2000) state that women play a prominent role in managing dairy animals. Finally, Burton & White (1984) mention specific animal-related tasks typically performed by women: bringing food and water to animals, milking and dairy production, meat preservation, caring for infant animals, and processing hides and wool. Therefore, this evidence suggests that women's involvement in agricultural jobs may also depend on the presence and type of animals on the farms.

1.2.4 Historically low female labour participation in agriculture

As these mechanisms do not seem to play a critical role in explaining the low female labour participation in Mexico's agricultural sector, the study discusses a strand of literature examining countries with historically low female labour participation in agriculture and potential explanations for this pattern. A growing body of studies has examined how distinct agricultural practices in pre-industrial societies shaped gender norms that constrained women's labour-market participation. The central argument is that gender norms often have their roots in early agrarian systems, which generated beliefs and attitudes that privileged men in the workplace while confining women to household chores. These studies explore how certain forms of agriculture have led to lower female involvement in agricultural tasks; however, their main contribution is demonstrating that these gender norms have had long-run implications for gender inequality observable today.

There is a theory on the use of the plough in pre-industrial societies. Boserup (1970) argues that the upper-body strength required to operate ploughs gave men a comparative advantage in this agricultural activity. Therefore, this theory distinguishes between two types of early agrarian societies: those cultivating plough-positive crops (e.g. wheat, teff, barley, rye) and those specialising in plough-negative crops (e.g. maize, sorghum, millet, roots, tree crops). The main argument is that in plough-based societies, men worked the fields while women remained in the household. Conversely, crops reliant on hand-held tools encouraged greater female labour participation on the farms. Over time, plough-based societies developed restrictive norms for women, while societies specialised in crops that did not require ploughing developed more egalitarian gender norms. Alesina et al. (2013) evaluated this hypothesis empirically and found that societies reliant on the plough display historically lower female participation in agricultural activities and

systematically lower FLPRs today.⁶ They also showed that individuals, ethnic groups, and countries with a legacy of plough use display higher gender inequality, as women report systematically lower participation in market work, entrepreneurship, and politics. They conclude that plough agriculture has left a lasting imprint on gender norms that persist to this day.

There is also a theory that examines the pace of the transition from hunting-gathering to agriculture. Hansen et al. (2015) argue that hunter-gatherer societies were relatively egalitarian because men and women had equally valuable economic roles: survival did not rely exclusively on men's hunting but rather on a diversified subsistence strategy in which women's gathering provided a steady and reliable calorie base. Nevertheless, the transition to agriculture altered this balance: men undertook strength-intensive farm tasks, while women were increasingly confined to domestic work and child-rearing. The authors argue that some hunter-gatherer societies transitioned faster to agriculture, thereby accumulating a 'longer agricultural history'. They argue that earlier transitions to agriculture signal greater technological advancement, which translates into higher fertility and a diminished role for women outside the home. Consequently, women were exposed to a patriarchal division of labour for longer, resulting in restrictive and deeply entrenched gender norms. Empirically, the results indicate that having a longer agricultural history is strongly associated with lower contemporary female labour force participation, as well as with other measures of gender equality. This finding is robust to an extensive set of potential confounders, including historical plough use.

Finally, there is also a study focusing on pastoralism. Becker (2025) argues that some pre-industrial societies were characterised by a greater reliance on pastoralism, a practice involving prolonged but 'non-permanent male absences' from their households to herd livestock to distant pasture grounds. The research contends that this practice diminished men's ability to monitor women's behaviour, generating elevated paternity uncertainty and incentivising the adoption of gender norms by limiting women's motivations for extramarital relations and thereby restricting female sexual autonomy. The author notes that pastoralist societies were more likely to exhibit generalised social disapproval of women interacting with men outside their families, thereby limiting their economic opportunities and their ability to participate freely in society. In contrast, the author highlights evidence from Teso (2019), who shows that 'permanent male absences' as a result of the transatlantic slave trade in Africa normalised women's work in traditionally male occupations, leading to higher female labour participation in those societies.

⁶ Alesina et al. (2013) argued that "historical plough use is associated with less female participation in agriculture historically and with less female participation in the labor force today."

Taken together, these studies suggest that various historical factors may also contribute to low female labour force participation in agriculture. My research finds evidence that Mexico is characterised by historically low female participation in agriculture. Nonetheless, identifying the specific factors behind this pattern lies beyond the scope of this study, as such an investigation would necessitate pre-modern data. Nonetheless, this is a fruitful direction for future research.

1.3 Methodology

This section outlines the methodology for examining the relationship between female labour participation and the characteristics of the local labour market structure where they live. Specifically, the section outlines the empirical strategy for analysing whether the percentage of jobs in agricultural and non-agricultural activities is associated with women's likelihood of participating in the workforce. The analysis draws on repeated cross-sectional microdata from the Mexican National Employment and Occupation Survey (ENOE) for 2005, 2010, 2015, and 2019. The empirical strategy is inspired by the model developed by Roncolato (2016), who employs probit regressions to estimate the probability of female labour force participation as a function of the percentage of non-agricultural jobs at the municipal level. This model specification is grounded in the structural transformation literature, where the reallocation of labour out of agriculture is always associated with a rise in GDP per capita (Herrendorf et al., 2014). The section begins with a brief description of the dataset, followed by an explanation of the model developed by Roncolato and a discussion of the adjustments that this study made to her methodology. The section concludes by presenting descriptive statistics outlining the principal attributes of the sample.

1.3.1 Dataset

This study uses micro-level data from the extended version of the ENOE household survey, conducted by Mexico's National Statistical Office (INEGI). The ENOE survey has served as the primary source of Mexico's labour market statistics since its inception in 2005, collecting employment data every quarter. Each dataset encompasses a sufficiently large sample size to represent both rural and urban areas across Mexico's 32 states. In addition, the surveys provide detailed information on respondents' labour status alongside socio-demographic variables, such as education level and marital status.

It is important to note that INEGI conducts an expanded survey in the first quarter of each year, whereas in the second, third and fourth quarters, it administers a basic survey. Hence, this study used four cross-sectional datasets derived from the ENOE surveys conducted in the first quarters of 2005, 2010, 2015 and 2019. The surveys from the first quarter were specifically chosen because they offer the most comprehensive information, as surveys from the other quarters exclude certain questions that are only included in the amplified survey. Moreover, the survey from the first quarter of 2019 was utilised because no survey was conducted in the first quarter of 2020 due to the COVID-19 pandemic. Aside from this exception, the analysis adheres to a five-year interval between the selected surveys. This timeframe is valuable because it captures changes in the sectoral distribution of employment not only across Mexican municipalities but also over time.

1.3.2 Econometric model

The econometric analysis used in this research is inspired by the model developed by Roncolato (2016). Her model examines whether women are more or less likely to participate in the workforce depending on the percentage of non-agricultural jobs at the municipal level. Hence, this variable can be used as a proxy for the economic structure of local labour markets, while it can also be considered a proxy of structural transformation at the municipal level. Her model also incorporates a set of control variables at multiple levels of analysis to address potential confounding factors correlated with women's likelihood of labour market participation. These controls operate at three distinct levels: individual characteristics (e.g. age, marital status, educational attainment), household characteristics (e.g. presence of children under six years of age), and municipal characteristics (e.g. percentage of households in the municipality with electricity). Hence, her model analyses the likelihood that women are participating in the workforce depending on the local labour market structure while controlling for individual, household and municipal characteristics.

The econometric model for this research, while inspired by Roncolato's modelling choice, also includes minor adjustments. The dependent variable remains a binary indicator for whether a woman is part of the labour force or not. The main independent variable remains the proportion of non-agricultural employment at the municipal level. However, my study expands its temporal and analytical scope. For instance, whereas Roncolato's analysis is based on a single year of data (2007), the present study draws on a repeated cross-sectional dataset from 2005, 2010, 2015, and 2019. Moreover, the study includes additional control variables that Roncolato did not include in her model.

Another minor difference is that Roncolato's analysis excluded students, the sick, and the disabled individuals from her sample, whereas this research includes them as part of the non-working population. This approach follows ILO guidelines, which classify students, the sick, the disabled, and retired individuals as part of the working-age population that is not participating in the labour force. Regarding age, this research restricts the sample to women aged 15-65. The lower bound of 15 corresponds to the legal working age in Mexico, the official age established by the ILO for estimating labour force participation rates, and the age used by Roncolato (2016). However, whereas the ILO includes all individuals aged 15 and above with no upper limit, both Roncolato and this research set an upper bound of 65. This upper limit of 65 years old corresponds to the legal retirement age in Mexico.

The research considered four model specifications to have an in-depth analysis of the relationship between female labour force participation and the sectoral composition of employment in local labour markets. All specifications share the same dependent variable, control variables and fixed effects. Therefore, the model specifications have an almost identical structure. However, each specification differs in how the core explanatory variable of that specific model captures the sectoral composition of local labour markets.

The first model specification uses the proportion of non-agricultural jobs at the municipal level as the core explanatory variable. The second model specification uses the share of jobs in agriculture, industry or services at the municipal level as the core explanatory variable. It should be noted that this model specification does not incorporate all three sectoral variables in a single regression. Instead, three distinct regressions are estimated. The third model specification includes the share of jobs in industry and services jointly in a single regression, omitting agriculture to avoid perfect multicollinearity. While Roncolato focuses on the share of non-agricultural jobs, this specification takes agriculture as the reference sector and allows the share of jobs in industry and services to vary on the right-hand side. This enables a more granular characterisation of structural transformation, as a labour market structure oriented to industry or services could have heterogeneous implications for female labour participation. Finally, the fourth model specification analyses the proportion of non-agricultural jobs at the municipal level and includes its squared term to capture potential non-linear relationships, as this is the original empirical strategy used by Roncolato (2016). These model specifications are expressed in the following equations:

First model specification

$$P(Y_i = 1) = \Phi(\beta_0 + \beta_1 \text{ShareNonAgriculture}_{m,t} + \sum_{j=5}^{j=2} \beta_j \text{CIC}_{i,t}^{(j)} + \sum_{j=13}^{j=6} \beta_j \text{CHC}_{h,t}^{(j)} + \sum_{j=24}^{j=14} \beta_j \text{CMC}_{m,t}^{(j)} + \mu_{s,t}) \quad (1)$$

Second model specification

$$P(Y_i = 1) = \Phi(\beta_0 + \beta_1 \text{ShareAgriculture}_{m,t} + \sum_{j=5}^{j=2} \beta_j \text{CIC}_{i,t}^{(j)} + \sum_{j=13}^{j=6} \beta_j \text{CHC}_{h,t}^{(j)} + \sum_{j=24}^{j=14} \beta_j \text{CMC}_{m,t}^{(j)} + \mu_{s,t}) \quad (2)$$

$$P(Y_i = 1) = \Phi(\beta_0 + \beta_1 \text{ShareIndustry}_{m,t} + \sum_{j=5}^{j=2} \beta_j \text{CIC}_{i,t}^{(j)} + \sum_{j=13}^{j=6} \beta_j \text{CHC}_{h,t}^{(j)} + \sum_{j=24}^{j=14} \beta_j \text{CMC}_{m,t}^{(j)} + \mu_{s,t}) \quad (3)$$

$$P(Y_i = 1) = \Phi(\beta_0 + \beta_1 \text{ShareServices}_{m,t} + \sum_{j=5}^{j=2} \beta_j \text{CIC}_{i,t}^{(j)} + \sum_{j=13}^{j=6} \beta_j \text{CHC}_{h,t}^{(j)} + \sum_{j=24}^{j=14} \beta_j \text{CMC}_{m,t}^{(j)} + \mu_{s,t}) \quad (4)$$

Third model specification

$$P(Y_i = 1) = \Phi(\beta_0 + \beta_1 \text{ShareIndustry}_{m,t} + \beta_2 \text{ShareServices}_{m,t} + \sum_{j=6}^{j=3} \beta_j \text{CIC}_{i,t}^{(j)} + \sum_{j=14}^{j=7} \beta_j \text{CHC}_{h,t}^{(j)} + \sum_{j=25}^{j=15} \beta_j \text{CMC}_{m,t}^{(j)} + \mu_{s,t}) \quad (5)$$

Fourth model specification

$$P(Y_i = 1) = \Phi(\beta_0 + \beta_1 \text{ShareNonAgriculture}_{m,t} + \beta_2 \text{ShareNonAgriculture}_{m,t}^2 + \sum_{j=6}^{j=3} \beta_j \text{CIC}_{i,t}^{(j)} + \sum_{j=14}^{j=7} \beta_j \text{CHC}_{h,t}^{(j)} + \sum_{j=25}^{j=15} \beta_j \text{CMC}_{m,t}^{(j)} + \mu_{s,t}) \quad (6)$$

The dependent variable in all specifications is denoted by ‘Y’, a binary variable indicating whether an individual (i) is participating in the labour force. It takes the value of 1 if the respondent is part of the labour force, 0 if they are not. ‘ Φ ’ is the standard normal cumulative distribution function of the probit model and ‘ β_0 ’ is the intercept.

The core explanatory variables change depending on the model specification. For instance, ‘*ShareNonAgriculture_{m,t}*’ constitutes the core explanatory variable in the first model specification. This variable captures the proportion of non-agricultural jobs relative to total employment in municipality (m) at time (t), expressed as a continuous variable ranging from 0 to 100. For instance, a value of 15 indicates that 85% of employment is in agriculture, reflecting a predominantly agricultural economy. Conversely, a value of 95 indicates that only 5% percent of jobs are in agriculture. Therefore, this variable serves as a proxy for the degree of structural transformation in the local labour market structure of each Mexican municipality.

The second model specification proceeds by estimating three separate regressions. Each regression captures the share of jobs in a specific economic sector—either in agriculture, industry, or services—in municipality (m) at time (t). The sum of these three variables is always equal to 100. For instance, a municipality might have 10% of jobs in agriculture, 20% in industry, and 70% in services. Hence, the model cannot include the three variables simultaneously in a single regression to avoid perfect multicollinearity. In this model specification, the first regression shows the correlation between the proportion of agricultural jobs at the municipal level and women’s likelihood of participating in the labour force. The second regression examines the relationship between the proportion of industrial jobs at the municipal level and female labour force participation. Finally, the third regression analyses how the proportion of service jobs in each municipality correlates with women’s participation in the labour market.

In the third model specification, the core explanatory variables are the share of industrial jobs at the municipal level, represented by ‘*ShareIndustry_{m,t}*’, as well as the share of jobs in services at the municipal level, represented by ‘*ShareServices_{m,t}*’. This model specification includes both variables in a single regression, omitting agriculture to avoid multicollinearity. Given that the literature on structural transformation focuses on the reallocation of labour out of agriculture, this specification includes these two variables simultaneously and shows how the percentage of jobs in industry and services, in municipality (m) at time (t), correlates with women’s likelihood of participating in the labour market.

Finally, in the fourth model specification, the core explanatory variables of the model are the share of non-agricultural jobs at the municipal level and its squared term. This model specification accounts for any potential non-linear relationships with women's likelihood of participating in the labour force. This final model specification is included in the analysis, as this is part of the empirical strategy of Roncolato (2016). By including this variable and its squared term, Roncolato finds that South Africa shows a U-shaped relationship between female labour force participation and the share of non-agricultural jobs at the municipal level. Her results indicate that women have a high probability of participating in the labour market if they live in municipalities where the vast majority of jobs are in agriculture. In contrast, she finds that in municipalities where roughly half the jobs are in agriculture, and the other half are non-agricultural, the probability that women are part of the labour force decreases. Finally, in municipalities where most of the jobs are non-agricultural, female labour participation rises again.

The coefficients of the core explanatory variables included in these model specifications will capture how the local labour market structure is correlated with female labour participation. In the first model specification, β_1 will show if there is a positive or negative relationship between female labour participation and the share of non-agricultural jobs at the municipal level. In the second model specification, β_1 will show whether the share of jobs in agriculture, industry or services is positively or negatively correlated with the probability that a woman will participate in the workforce. As noted above, the sectoral employment shares at the municipal level are not included jointly in a single regression but are instead estimated in three separate probit regressions. In the third model specification, β_1 and β_2 will show the relationship between the share of jobs in industry and services, respectively. In this equation, the share of jobs in agriculture is omitted to keep it as the base sector, while the other two sectors (industry and services) are included simultaneously in a single regression, allowing them to vary on the right-hand side of the equation. Finally, in the fourth model specification, β_1 and β_2 will show the relationship between the share of non-agricultural jobs and its squared term, in order to examine whether there is a non-linear relationship with women's likelihood of female labour participation in Mexico.

As previously explained, the four model specifications share the same control variables. 'CIC' represents a vector of four control variables that account for the individual characteristics of each respondent in the sample. This vector includes 'age' and 'age squared' as the relationship between female labour participation and age is often non-linear. The model further controls for marital status and educational attainment, with the latter capturing the highest level of education achieved by the respondents.

‘CHC’ represents a vector of eight control variables capturing the characteristics of each household (h). These variables include: 1) socioeconomic stratum of the household; 2) number of children under five years old in the household; 3) number of household members; 4) sex of the household head; 5) age of the household head; and 6) educational attainment of the household head. Furthermore, the econometric model includes two variables that describe the characteristics of the household’s location: 7) population size of the location of the household; and 8) whether the household is in a rural or urban area of Mexico.

‘CMC’ represents a vector of 11 control variables at the municipal level, which are discussed in detail in the next subsection. These variables are employed to account for different characteristics of the municipality where each respondent lives. These control variables were estimated using the sampling weights provided by INEGI; their purpose is to examine whether the relationship between the dependent variable and the core explanatory variables is statistically significant after controlling for other municipal characteristics.

Finally, μ represents the fixed effects incorporated into the model to control for unobserved heterogeneity across time (t) and across Mexican states (s), using Mexico City and the first quarter of 2005 as the base categories. Therefore, $i \in \{1, \dots, N\}$ denotes the index for each individual in the sample, $h \in \{1, \dots, H\}$ represents each household in the sample, $m \in \{1, \dots, M\}$ represents each municipality in the sample, $t \in \{2005 \text{ 1Q}, 2010 \text{ 1Q}, 2015 \text{ 1Q}, 2019 \text{ 1Q}\}$ represents the year and quarter of each of the surveys considered in this study, and $s \in \{1, \dots, 32\}$ is an index for the 32 states in Mexico.

1.3.3 Variable description and summary statistics

This subsection presents a detailed description of the different variables employed in the model. It explains the dependent variable, the key explanatory variables and the control variables that capture individual, household and municipal characteristics. Each variable is defined and discussed with reference to its construction and the anticipated relationship with female labour force participation, drawing on relevant theoretical and empirical contributions. The descriptive statistics of all these variables are reported in Table 1.1, included at the end of this section. Finally, a set of charts in Figure 1.16 presents patterns of female labour participation considering some of the control variables that were included in the model (see Appendix B, p. 64).

1.3.3.1 Dependent variable

According to the ILO and INEGI, the labour force participation rate is measured by considering individuals aged 15 and older. This measure captures the proportion of the working-age population that is either employed or actively seeking employment. In line with this convention, the model's dependent variable is a binary indicator of whether an individual is part of the labour force. This variable takes the value of 1 if the respondent is employed or actively seeking work, and 0 if they are not in the workforce. Individuals enrolled in school, engaged in unpaid domestic work, retired, disabled, or unable to work are classified as the non-working population. This approach aligns with international standards and enables a granular analysis of the determinants of female labour participation using individual-level data.

1.3.3.2 Explanatory variables

The key explanatory variable in the model is the proportion of non-agricultural employment at the municipal level, expressed as a continuous variable ranging from 0 to 100. A higher value of this variable (such as 80%) indicates that non-agricultural activities dominate the local employment structure, reflecting a more diversified economy. In contrast, lower values (such as 10%) suggest that agricultural employment predominates, pointing to a more rural or agrarian economic context. This variable is grounded in the structural transformation literature, which tends to interpret the reallocation of labour from agriculture to non-agricultural activities as a defining feature of economic development. Perkins et al. (2013) explained that, at the lowest levels of income per capita, agriculture dominates both in terms of GDP share and total employment. Conversely, as industry and services expand, the share of agriculture in GDP and total employment diminishes.

The first model specification uses the proportion of non-agricultural jobs at the municipal level as the core explanatory variable. This will show how labour reallocation away from agriculture relates to women's likelihood of participating in the labour force. The next model specifications provide a more granular analysis by disaggregating non-agricultural employment into industry and services. This enables a nuanced assessment of how female labour participation in Mexico varies depending on sector-specific employment structures.

1.3.3.3 Control variables

This subsection explains the control variables incorporated into the econometric model, as they may be related to female labour participation. These variables encompass respondents' individual attributes, household characteristics, and key features of their living environment. Figure 1.16 presents various charts depicting the relationship between female labour participation and specific control variables examined in this section (see Appendix B, p. 64).

The initial individual-level control variables are age and age squared, given the non-linear relationship between labour status and age. The model also includes educational attainment as a categorical variable indicating the highest level achieved. Research shows that the relationship between women's education and labour market participation varies across countries, precluding assumptions of a universally positive, linear relationship. Klasen et al. (2021) provide micro-level evidence on these disparities across eight developing countries, finding U-shaped patterns in some nations and linear patterns in others. Figure 1.16 shows that Mexican women with low education levels have the lowest labour force participation rates, while highly educated women have the highest. Therefore, educational attainment is included as a control variable, with an expected positive and linear relationship with female labour participation in Mexico.

Other control variables are marital status and number of children. Surprisingly, Figure 1.16 shows that married women in Mexico have higher labour force participation rates than single women. Additionally, women without children have lower FLPRs than women with children. This is counterintuitive, as single women and women without children typically participate more in the workforce as they have fewer caregiving duties. Nevertheless, the sample considers young women starting at age 15, as this is both the minimum legal working age in Mexico and the lower bound of the working-age population according to ILO. This leads to many young women without children being classified as working-age population that is not participating in the labour market, as many of them are studying rather than working. As a result, single women and women without children show lower labour force participation rates than wives and women with children.

The econometric model also includes a control variable that shows if the respondent is living in a household with one of these four socioeconomic strata: low, medium-low, medium-high, or high. According to INEGI (2020), this variable is derived using multivariate statistical methods based on 34 indicators assessing both the economic status of individuals and the physical attributes and amenities of their households. Key indicators include access to medical services, educational attainment, literacy, household flooring type, overcrowding, access to electricity, water

and drainage, and ownership of goods such as televisions, cars, mobile phones, refrigerators and washing machines. Hence, the ‘socioeconomic stratum’ variable serves as a proxy for the financial situation of household members, as this may influence a woman’s decision to engage in economic activities. Verick (2014) argues that poor women in low-income countries are more likely to participate in the labour market, often in subsistence activities or informal employment. Conversely, Lampietti and Stalker (2000) found that, in six of nine Latin American countries analysed, poor women exhibited lower labour force participation rates than their non-poor counterparts. Figure 1.16 shows that, in Mexico, women from low socioeconomic strata have the lowest labour participation rates.

López-Acevedo et al. (2021) estimate that residing in a Mexican urban household is associated with an 11.1 percentage point increase in women's probability of employment. Hence, the model includes a control variable equal to 1 for urban households and to 0 for rural households. Falk and Leoni (2010) also found that population density is positively associated with FLPRs, suggesting that densely populated areas offer a wider and more diverse array of employment opportunities for women. Consequently, a population density variable was incorporated, coded as 1 for households located in areas with more than 100,000 inhabitants, 2 for those in areas with between 15,000 and 99,000 inhabitants, 3 for those in areas between 2,500 and 14,999 inhabitants, and 4 for those in areas with fewer than 2,500 inhabitants. The regression analysis also includes additional household-level controls: the age, gender and educational attainment of the household head, the number of household members, and the number of children under five years of age.

Finally, the econometric model incorporates control variables to capture different characteristics of the municipality. Including these controls in the regression analysis is crucial because the key explanatory variable in this empirical analysis is the share of non-agricultural jobs at the municipal level. Thus, the analysis accounts for other characteristics of the municipality where the respondents reside. This approach facilitates a more accurate assessment of whether there is a statistically significant relationship between a woman’s likelihood of participating in the workforce and the sectoral employment distribution in her municipality.

The control variables at the municipal level are: 1) average age of women in the municipality; 2) proportion of women in the municipality with elementary schooling or less; 3) proportion of women in the municipality with secondary schooling; 4) proportion of women in the municipality with high school education; 5) proportion of single women in the municipality; 6) proportion of married women in the municipality; 7) proportion of women in the municipality who are in a free union relationship; 8) proportion of people in the municipality from a low

socioeconomic stratum; 9) proportion of people in the municipality from a medium-low socioeconomic stratum; 10) average number of sons or daughters among women between 20 and 35 years old (used as a proxy of the fertility rate in the municipality); and 11) proportion of people in the municipality who have migrated from their city or locality to retain or obtain their current job.

Finally, it is worth noting that this study includes a control variable that captures the proportion of the municipality's population who have migrated from their city or locality to obtain or maintain their current job. This is particularly relevant because within-country migration is generally easier than cross-country migration, which may raise concerns that local labour market structures are being influenced by internal migration not captured by the model. In fact, Roncolato (2016) mentioned that her model does not control for internal migration, since the South African dataset does not contain this information. Hence, including this variable in the regression analysis helps to assess the robustness of the results after accounting for the relative ease of internal migration compared with the greater difficulty of migrating across countries.

To conclude this section, Table 1.1 presents the descriptive statistics of all the variables considered in the regression models.

Table 1.1 – Descriptive statistics

	Women						Men					
	Obs.	%	Mean	Std. Dev.	Min	Max	Obs.	%	Mean	Std. Dev.	Min	Max
<i>Dependent variable</i>												
Participating in the labour force	539,288	-	0.49	0.500	0	1	486,493	-	0.83	0.373	0	1
<i>Main independent variables</i>												
Share of non-agricultural jobs	539,287	-	89.90	16.701	0	100	486,492	-	89.81	16.881	0	100
Share of agricultural jobs	539,287	-	10.10	16.701	0	100	486,492	-	10.19	16.881	0	100
Share of industrial jobs	539,287	-	25.12	10.355	0	100	486,492	-	25.25	10.446	0	100
Share of service jobs	539,287	-	64.21	16.591	0	100	486,492	-	63.97	16.689	0	100
Share of unspecified jobs	539,287	-	0.57	1.367	0	27.42	486,492	-	0.60	1.414	0	27.42
<i>Individual characteristics</i>												
Age	539,288	-	36.59	13.630	15	65	486,493	-	35.96	13.748	15	65
<i>Marital status</i>												
Free union	73,730	13.67	-	-	-	-	69,159	14.22	-	-	-	-
Separated	26,023	4.83	-	-	-	-	11,633	2.39	-	-	-	-
Divorced	12,068	2.24	-	-	-	-	5,312	1.09	-	-	-	-
Widowed	19,176	3.56	-	-	-	-	3,537	0.73	-	-	-	-
Married	242,502	44.97	-	-	-	-	223,263	45.89	-	-	-	-
Single	165,751	30.74	-	-	-	-	173,552	35.67	-	-	-	-
Missing values	38	0.01	-	-	-	-	37	0.01	-	-	-	-
Total	539,288	100					486,493	100	-	-	-	-
<i>Level of education</i>												
No studies	23,087	4.28	-	-	-	-	15,442	3.17	-	-	-	-
Primary school	130,380	24.18	-	-	-	-	108,351	22.27	-	-	-	-
Secondary school	144,116	26.72	-	-	-	-	139,296	28.63	-	-	-	-
High school	101,127	18.75	-	-	-	-	105,662	21.72	-	-	-	-
Technical school	44,941	8.33	-	-	-	-	17,537	3.6	-	-	-	-
Undegraduate	89,099	16.52	-	-	-	-	92,665	19.05	-	-	-	-
Postgraduate	6,276	1.16	-	-	-	-	7,201	1.48	-	-	-	-
Missing values	262	0.05					339	0.07				
Total	539,288	100					486,493	100				

	Women						Men					
	Obs.	%	Mean	Std. Dev.	Min	Max	Obs.	%	Mean	Std. Dev.	Min	Max
<i>Household characteristics</i>												
Total household members	539,288	-	4.57	1.945	1	11	486,493	-	4.55	1.952	1	11
Number of kids in the household	539,288	-	0.45	0.726	0	4	486,493	-	0.41	0.699	0	4
Located in urban/rural area	539,288	-	0.62	0.484	0	1	486,493	-	0.62	0.485	0	1
<i>Population size of household location</i>												
More than 100,000 inhabitants	330,205	61.23	-	-	-	-	296,308	60.91	-	-	-	-
Between 15,000 and 99,999	64,692	12	-	-	-	-	57,891	11.9	-	-	-	-
Between 2,500 and 14,999	64,544	11.97	-	-	-	-	57,734	11.87	-	-	-	-
Less than 2,500 inhabitants	79,847	14.81	-	-	-	-	74,560	15.33	-	-	-	-
<i>Total</i>	539,288	100					486,493	100				
<i>Socioeconomic strata</i>												
Low	71,245	13.21	-	-	-	-	65,719	13.51	-	-	-	-
Medium-low	263,097	48.79	-	-	-	-	240,300	49.39	-	-	-	-
Medium-high	146,922	27.24	-	-	-	-	129,864	26.69	-	-	-	-
High	58,024	10.76	-	-	-	-	50,610	10.4	-	-	-	-
<i>Total</i>	539,288	100					486,493	100				
<i>Household head characteristics</i>												
Age of the household head	539,288	-	48.01	13.404	15	98	486,493	-	47.34	13.114	15	98
Sex of the household head	539,288	-	0.29	0.452	0	1	486,493	-	0.16	0.367	0	1
<i>Household head education</i>												
No studies	34,503	6.4	-	-	-	-	29,707	6.11	-	-	-	-
Primary school	179,252	33.24	-	-	-	-	160,197	32.93	-	-	-	-
Secondary school	127,822	23.7	-	-	-	-	118,981	24.46	-	-	-	-
High school	70,299	13.04	-	-	-	-	66,763	13.72	-	-	-	-
Technical school	32,651	6.05	-	-	-	-	24,484	5.03	-	-	-	-
Undgraduate	81,915	15.19	-	-	-	-	75,873	15.6	-	-	-	-
Postgraduate	10,048	1.86	-	-	-	-	9,005	1.85	-	-	-	-
Missing values	2,798	0.52	-	-	-	-	1,483	0.3	-	-	-	-
<i>Total</i>	539,288	100	-	-	-	-	486,493	100	-	-	-	-

	Women						Men					
	Obs.	%	Mean	Std. Dev.	Min	Max	Obs.	%	Mean	Std. Dev.	Min	Max
<i>Municipal characteristics</i>												
Share of people who migrated to keep or maintain their current job	538,443	-	2.68	3.589	0	100	486,484	-	2.72	3.712	0	100
Share of women in the municipality with primary school or less	539,288	-	32.94	14.057	0	100	486,493	-	32.88	14.038	0	100
Share of women in the municipality with secondary school	539,288	-	26.53	6.570	0	100	486,493	-	26.67	6.604	0	100
Share of women in the municipality with high school	539,288	-	16.68	5.429	0	60.00	486,493	-	16.70	5.449	0	60.00
Share of women in the municipality that are single	539,288	-	30.06	5.321	0	66.67	486,493	-	29.81	5.402	0	66.67
Share of women in the municipality that are married	539,288	-	43.37	7.640	0	100	486,493	-	43.53	7.712	0	100
Share of women in the municipality that are in a free-union relationship	539,288	-	12.44	6.702	0	100	486,493	-	12.57	6.786	0	100
Share of people in the municipality from a low socioeconomic stratum	539,288	-	13.98	26.299	0	100	486,493	-	13.97	26.312	0	100
Share of people in the municipality from a medium-low socioeconomic stratum	539,288	-	49.18	24.302	0	100	486,493	-	49.11	24.298	0	100
Average children per woman aged 20-35 in the municipality	539,288	-	1.29	0.339	0	4.33	486,493	-	1.29	0.342	0	4.33
Average age of women in the municipality	539,288	-	39.48	2.667	26.67	59.79	486,493	-	39.49	2.684	26.67	59.79
<i>Year/quarter</i>												
1st quarter of 2005	133,630	24.78	-	-	-	-	118,383	24.33	-	-	-	-
1st quarter of 2010	134,126	24.87	-	-	-	-	120,816	24.83	-	-	-	-
1st quarter of 2015	134,534	24.95	-	-	-	-	122,246	25.13	-	-	-	-
1st quarter of 2019	136,998	25.40	-	-	-	-	125,048	25.70	-	-	-	-

	Women						Men					
	Obs.	%	Mean	Std. Dev.	Min	Max	Obs.	%	Mean	Std. Dev.	Min	Max
<i>Mexican states</i>												
Aguascalientes	18,621	3.04	-	-	-	-	16,654	3.02	-	-	-	-
Baja California	20,117	3.28	-	-	-	-	19,521	3.54	-	-	-	-
Baja California Sur	12,934	2.11	-	-	-	-	12,767	2.32	-	-	-	-
Campeche	17,275	2.82	-	-	-	-	15,832	2.87	-	-	-	-
Coahuila	20,106	3.28	-	-	-	-	18,958	3.44	-	-	-	-
Colima	16,455	2.68	-	-	-	-	14,986	2.72	-	-	-	-
Chiapas	22,045	3.6	-	-	-	-	19,081	3.46	-	-	-	-
Chihuahua	18,568	3.03	-	-	-	-	17,065	3.1	-	-	-	-
Mexico City	16,401	2.67	-	-	-	-	14,251	2.59	-	-	-	-
Durango	18,496	3.02	-	-	-	-	16,138	2.93	-	-	-	-
Guanajuato	26,183	4.27	-	-	-	-	22,965	4.17	-	-	-	-
Guerrero	18,846	3.07	-	-	-	-	16,310	2.96	-	-	-	-
Hidalgo	16,781	2.74	-	-	-	-	14,172	2.57	-	-	-	-
Jalisco	22,315	3.64	-	-	-	-	20,251	3.68	-	-	-	-
México	28,042	4.57	-	-	-	-	25,199	4.58	-	-	-	-
Michoacán	19,140	3.12	-	-	-	-	16,516	3	-	-	-	-
Morelos	17,190	2.8	-	-	-	-	14,901	2.71	-	-	-	-
Nayarit	17,471	2.85	-	-	-	-	15,975	2.9	-	-	-	-
Nuevo Leon	20,576	3.36	-	-	-	-	19,998	3.63	-	-	-	-
Oaxaca	21,928	3.58	-	-	-	-	18,273	3.32	-	-	-	-
Puebla	23,516	3.84	-	-	-	-	20,292	3.68	-	-	-	-
Querétaro	18,151	2.96	-	-	-	-	16,062	2.92	-	-	-	-
Quintana Roo	14,513	2.37	-	-	-	-	14,339	2.6	-	-	-	-
San Luis Potosi	18,786	3.06	-	-	-	-	16,693	3.03	-	-	-	-
Sinaloa	17,989	2.93	-	-	-	-	16,683	3.03	-	-	-	-
Sonora	16,587	2.71	-	-	-	-	15,813	2.87	-	-	-	-
Tabasco	18,414	3	-	-	-	-	16,271	2.95	-	-	-	-
Tamaulipas	19,506	3.18	-	-	-	-	17,870	3.25	-	-	-	-
Tlaxcala	18,188	2.97	-	-	-	-	15,925	2.89	-	-	-	-
Veracruz	21,472	3.5	-	-	-	-	18,084	3.28	-	-	-	-
Yucatán	17,409	2.84	-	-	-	-	16,029	2.91	-	-	-	-
Zacatecas	19,133	3.12	-	-	-	-	16,807	3.05	-	-	-	-

1.4 Results

This section presents the empirical findings from the analysis estimating the likelihood that Mexican women will participate in the workforce, conditional on the sectoral distribution of employment in their municipality of residence. The analysis draws on repeated cross-sectional data from the first quarters of 2005, 2010, 2015 and 2019, obtained from Mexico's official labour force surveys. The dependent variable is a binary indicator for whether a woman is part of the workforce or not. The key explanatory variable is the share of non-agricultural jobs at the municipal level. The econometric analysis employs probit regressions with probability weights, clustered standard errors at the municipal level, and includes state and time fixed effects to control for unobserved heterogeneity. The discussion of the regression results should be interpreted as descriptive evidence, so the reader should avoid inferring causal claims.

Table 1.2 presents the regression results. Column (1) shows a positive, statistically significant relationship between the share of non-agricultural employment and female labour force participation. Column (2) shows the corresponding negative relationship with the agricultural employment share. These results show the same coefficient, differing only in sign, because the two shares sum to 100%. Thus, the results suggest that women are significantly less likely to participate in the workforce if they live in municipalities where agricultural employment predominates.

It is important to highlight Mexico's broad definition of labour force participation, which includes subsistence farming, agriculture for self-consumption, and unpaid work on family farms. Anyone reporting at least one hour of economic activity per week on a farm is classified as employed. Nevertheless, despite this permissive classification, many women in agricultural regions report no economic activity whatsoever, indicating that survey underestimation is unlikely to account for the low female labour force participation observed in Mexico's agricultural regions.⁷

It is also important to emphasise that Mexico's and South Africa's labour market structures share certain similarities. According to World Bank data from 2022, South Africa had 19% of all

⁷ The ENOE questionnaire starts with a comprehensive series of questions to identify whether an individual is part of the labour force. The initial question asks: "Did you work for at least one hour last week?" If the respondent answers negatively, INEGI continues to probe in order to verify whether the person has genuinely not worked. The subsequent questions include: "Regardless of what you just told me, did you spend at least one hour last week on an activity that provided you with income?" Then, "Regardless of what you just told me, did you spend at least one hour last week helping with the land or business of a family member or another person?" A further question asks: "Although you already told me that you did not work last week, do you have any job, business, or engage in any activity on your own account?" These questions form part of INEGI's methodology in the ENOE survey to ensure that an individual is genuinely not part of the labour force.

jobs in agriculture, 18% in industry, and 63% in services. Meanwhile, Mexico had 13% in agriculture, 25% in industry, and 62% in services. While not perfectly comparable, this is not a comparison of Mexico with a country where agricultural jobs predominate or where industrial employment is negligible. Both countries have service sectors of nearly identical size, while Mexico has a slightly higher share of jobs in industry and South Africa somewhat more in agriculture. So even when they do not have exactly the same labour-market structure, they are two upper-middle-income economies, both highly unequal, yet reporting substantially different patterns of female participation in agricultural labour markets.

The strikingly low participation of Mexican women in agriculture is also evident from different perspectives. The scatterplots presented in Figure 1.4 (p. 51) show that FLPRs are lower in agricultural regions of the country at both the state and the municipal levels. Furthermore, Figure 1.5 (p. 52) shows that, in 2019, around 90% of agricultural workers in Mexico were men with only 10% being women. Additionally, Figure 1.6 (p. 52) shows that, among the female workforce, 3% were employed in agriculture, 17% in industry, and 80% in services. Taken together, this evidence shows that very few women in Mexico are engaged in agricultural activities.

The research also analyses, separately and simultaneously, the relationship between female labour force participation and the proportion of jobs in industry and services at the municipal level. This strategy captures how sector-specific employment shares correlate with women's labour-market participation. Columns (3) and (4) indicate that when non-agricultural jobs are disaggregated into industry and services, both sectors show a positive relationship with female labour participation. Furthermore, Column (5) shows that, when industry and services are included together in the same regression, the results remain positive and statistically significant. The results also indicate that a labour market oriented towards services is associated with higher FLPRs than one oriented towards industry. In addition, labour markets with a greater percentage of industrial jobs are associated with higher FLPRs than those where agriculture predominates.

The empirical strategy followed by Roncolato (2016) also includes a squared term for the percentage of non-agricultural jobs at the municipal level to capture any potential non-linear relationships. Following this methodology, she finds that female labour force participation is high in agricultural municipalities and in those where non-agricultural employment predominates, but declines in municipalities where there is no clear predominance of either agricultural or non-agricultural jobs. Hence, she discovers that, within South Africa, there is a U-shaped relationship between female labour force participation and the proportion of non-agricultural jobs.

Accordingly, this analysis includes an additional specification with a squared term for the percentage of non-agricultural jobs to capture any non-linear relationship. The results are presented in Column (6) of Table 1.2. For ease of interpretation, Figure 1.1 plots the predicted probabilities of labour-market participation for men and women against the municipal percentage of non-agricultural jobs. The figure illustrates that, in municipalities where more than 70% of jobs are non-agricultural, women's likelihood of participating in the workforce is above 60%. However, in municipalities where more than 90% of jobs are in agriculture, women's probability of labour market participation declines to 30%–40%.

On the other hand, men's predicted probabilities of participation in the workforce remain above 90%, regardless of the local labour market structure. The regression results for men are presented in Table 1.4 (see Appendix B, p. 48). In addition, the average marginal probabilities for men and women are presented in Table 1.5 (see Appendix B, p. 49). Overall, the results suggest that labour-market structure is not associated with significant variations in men's likelihood of labour force participation, while, for women, there are substantial variations. Women residing in non-agricultural regions in Mexico are more likely to participate in the workforce. Women living in agricultural regions, however, are less likely to participate in the labour market and are consequently more likely to remain at home.

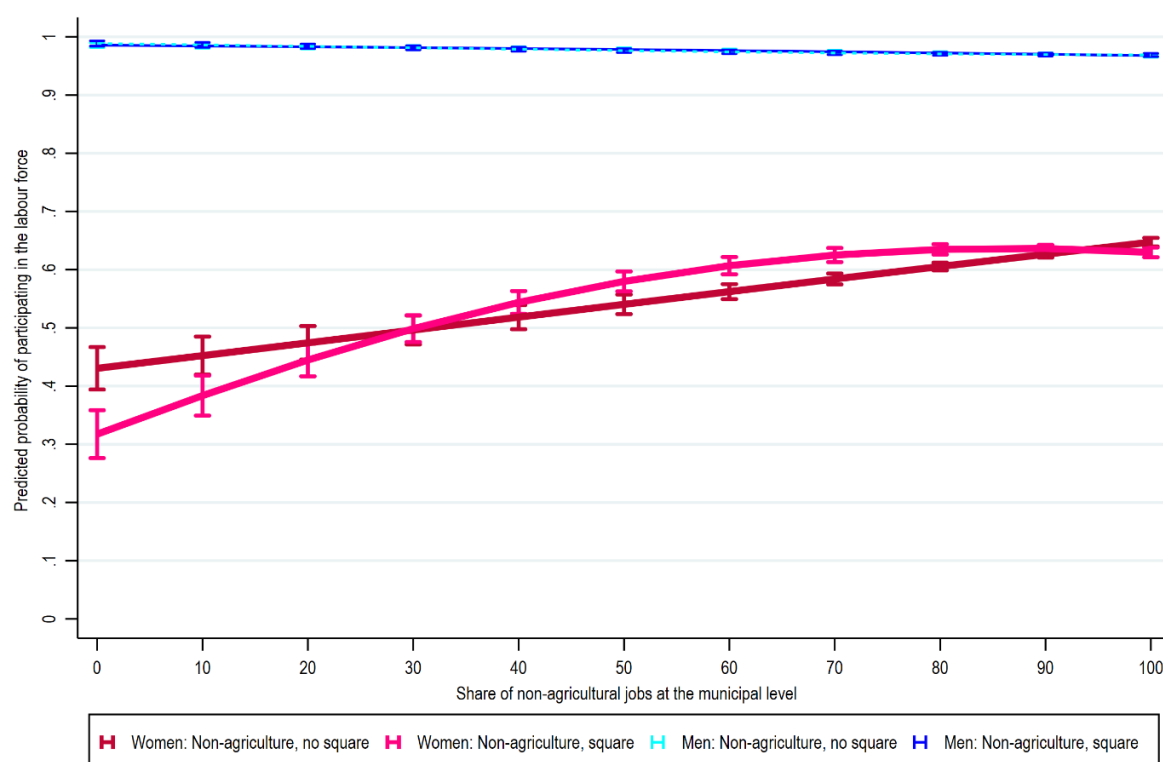
It bears emphasis that these results are strictly correlational and therefore do not permit any causal interpretation. Moreover, the absence of certain control variables may raise concerns about omitted variable bias. Structural conditions such as organised crime and violence constitute one potential source of bias, while wage premiums that vary by sector or municipality may also be another source of endogeneity if they shape the allocation of workers across activities. To address these concerns, this paper reports the selection ratio proposed by Altonji et al. (2005), who proposed using selection on observables as a benchmark for assessing the relevance of selection on unobservables (see Table 1.6 in Appendix B, p. 50). This ratio quantifies how much stronger selection on unobserved determinants would need to be, relative to selection on observed controls, to attribute the estimated results entirely to omitted variable bias. The estimated selection ratios reported in this paper suggest that unobserved selection would need to be implausibly large—at least comparable to, and in several cases substantially stronger than—selection on observables to overturn the core findings. This provides reassurance that the estimated correlations between different local labour market structures and women's likelihood of participating in the labour force are not expected to be reversed by the inclusion of additional controls.

Table 1.2 – Results of probit regressions: likelihood that Mexican women are participating in the labour force, depending on the sectoral distribution of employment at the municipal level.

	(1) Non-Agricultural	(2) Agriculture	(3) Industry	(4) Services	(5) Industry & services	(6) Non-Agricultural (Squared)
Share of non-agricultural jobs at the municipal level	0.00553*** (0.00053)					0.01900*** (0.00170)
Share of agricultural jobs at the municipal level		-0.00553*** (0.00053)				
Share of industrial jobs at the municipal level			0.00223*** (0.00058)		0.00502*** (0.00063)	
Share of service jobs at the municipal level				0.00388*** (0.00056)	0.00612*** (0.00060)	
Share of non-agricultural jobs at the municipal level (squared)						-0.00011*** (0.00001)
Control variables - Individual	Yes	Yes	Yes	Yes	Yes	Yes
Control variables - Household	Yes	Yes	Yes	Yes	Yes	Yes
Control variables - Municipal	Yes	Yes	Yes	Yes	Yes	Yes
Year/quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Clustered standard errors	Yes	Yes	Yes	Yes	Yes	Yes
Constant	-4.58129*** (0.22131)	-4.02813*** (0.20358)	-3.71910*** (0.19850)	-4.21933*** (0.20985)	-4.64716*** (0.22116)	-4.86167*** (0.21011)
Observations	538,433	538,433	538,433	538,433	538,433	538,433

Notes
Base category for state fixed effects: Mexico City
Base category for time fixed effects: First quarter of 2019.
Clustered standard errors at the municipal level.
Standard errors in parentheses
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Figure 1.1 - Predicted probabilities that women and men participate in the labour force, depending on the share of non-agricultural jobs in the municipality where they live.



Note: The predicted probabilities to plot these estimations are presented in Table 1.5 (see Appendix B, p. 49).

1.5 In-depth exploratory analysis

Why is female labour participation so low in the Mexican agricultural sector? To address this question, this section first outlines key features of the sector and discusses sharp contrasts across regions. Once the agricultural context is established, the analysis proceeds to examine alternative mechanisms that may explain women's limited participation in agricultural work. Lastly, the section concludes by analysing historical trends in female labour force participation and the process of structural transformation in Mexico between 1900 and 2024.

The current structure of Mexico's agricultural labour markets reflects pronounced regional contrasts. The southern states exhibit the highest proportion of agricultural jobs, while the northern states sharing borders with the USA tend to report the highest proportion of industrial jobs (see Figure 1.7 in Appendix B, p. 53). Furthermore, the data show that the average farm size, measured in hectares, is considerably smaller in the southern states, whereas the northern states tend to have large-scale agricultural units (see Figure 1.8 in Appendix B, p. 55). Moreover, agricultural processes in the southern states rely mostly on self-employment and unpaid family labour, while the northern states depend primarily on *jornaleros*, daily wage workers typically hired under informal arrangements (see Figure 1.9 in Appendix B, p. 57). Despite these clear regional differences, agricultural activities in Mexico show a consistent pattern: they rely on different types of informal labour. The figures illustrate that northern states bordering the USA are characterised by large-scale agriculture and extensive mechanisation, but they rely on casual wage workers hired informally. Meanwhile, the poorest southern states are characterised by small farm sizes, low levels of mechanisation, and a predominance of self-employment and unpaid family labour (see Table 1.9 in Appendix B, p. 58).

After presenting an overview of agricultural labour markets in Mexico, the discussion turns to explore different mechanisms that may explain the low engagement of Mexican women in agricultural activities. First, the supply-side mechanism suggests that the remittances sent by Mexicans working in the USA may be reducing the labour supply of women in rural areas of the country. Figure 1.10 (p. 59) shows the relationship between remittances received in Mexican municipalities and the share of women participating in the agricultural workforce of those municipalities. Nevertheless, the figure does not show explanatory power to conclude that more remittances are associated with a lower percentage of women in the agricultural workforce at the municipal level.

Second, the demand-side mechanism suggests that low female labour participation in agriculture might be the result of the adoption of agricultural machinery that has reduced the demand for female workers in the sector. Figure 1.11 in Appendix B (p. 60) shows the relationship between agricultural mechanisation (proxied by tractor use) and women's participation in agricultural labour at the municipal level. Municipalities in northern Mexican states bordering the USA exhibit widespread tractor usage, while municipalities in the poorest southern states show minimal to non-existent mechanisation. Nevertheless, the figure shows no association between tractor usage and female agricultural employment.

Third, the animal-draught mechanism suggests that the use of draught animals on farms is associated with lower female labour participation in agricultural activities. Boserup (1970) argued that draught animals reduce the demand for weeding—a task traditionally performed by women—while land preparation using draught animals requires upper-body strength, making it predominantly a male activity. Figure 1.12 (p. 60) explores this relationship by plotting the percentage of agricultural units using draught or yoked animals against female labour participation in the agricultural workforce of Mexican municipalities. Surprisingly, the figure shows a positive relationship: female labour participation in agriculture is higher in municipalities with more farms using draught and yoked animals. As noted by Burton & White (1984) women play prominent roles in performing animal-related tasks such as feeding and caring for livestock. Therefore, this positive correlation may reflect women's involvement in these complementary livestock activities.

Based on this evidence, this research argues that low female labour participation in agriculture is not a recent phenomenon driven by remittances, mechanisation, or the use of draught animals. Instead, it represents a historical pattern. After analysing decennial population census data, it is discovered that Mexico has had low levels of female participation in agriculture since the beginning of the twentieth century. Figure 1.13 in Appendix B (p. 61) shows the share of women in Mexico's agricultural workforce between 1900 and 2024 using historical data collected at the start of each decade. The figure indicates that the share of women in the agricultural workforce has remained consistently low for more than 125 years. Hence, the data points to a long-standing predominance of male workers in Mexico's agricultural sector.

Moreover, Figure 1.14 in Appendix B (p. 62) illustrates the process of structural transformation in Mexico from 1900 to 2024. The historical data were also obtained from Mexico's decennial population census. The figure shows that, in the early decades of the twentieth century, Mexico was predominantly agrarian, with some 65%–70% of the labour force employed in the primary sector. The data also shows that Mexico's structural transformation began to unfold in the

1940s. A discernible shift emerged in the sectoral composition of employment, driven by the progressive expansion of industry and services, which gradually assumed a more prominent role in the labour market.

Nevertheless, Sen (2023) defines a structurally developed economy as one in which the industrial sector employs a greater proportion of the labour force than the agricultural sector. Figure 1.14 in Appendix B (p. 62) shows that Mexico reached this stage of structural transformation in 1990. By this time, the service sector had become the dominant source of employment, the industrial sector was the second largest, and agriculture was relegated to third place in the country's employment structure. Hence, Mexico was predominantly agrarian for nearly the entire twentieth century, only achieving structural development in 1990. Since then, the industrial sector has consistently employed around 25% of Mexico's labour force. Over the same period, agricultural employment continued its long-run decline, while the service sector experienced uninterrupted expansion, ultimately accounting for nearly two-thirds of total employment in the Mexican economy by 2024.

Historical data also show that FLPRs in Mexico followed a U-shaped pattern during the first half of the twentieth century (see Figure 1.15 in Appendix B, p. 63). In 1900, roughly 19% of women were participating in the labour market. This figure declined in subsequent decades, reaching its lowest level in 1930, when only about 5% of working-age women were participating in the workforce. By 1960, FLPRs had increased to nearly 20%, comparable with levels observed at the start of the century. Hence, the period from 1900 to 1960 is best characterised as a U-shaped pattern in the historical trajectory of FLPRs in Mexico. Since then, FLPRs have been increasing consistently, signalling a long-term upward trend that is likely to continue in the coming decades.

This study has examined Mexico's structural transformation and female labour participation since 1900, revealing that women's minimal involvement in agriculture is a longstanding historical pattern rather than a recent development. The proportion of Mexican women in the agricultural workforce has remained consistently low for over a century. Several characteristics of early agrarian societies in Mexico may explain the reasons behind this pattern. Nonetheless, the precise factors, mechanisms or gender norms sustaining this historical pattern remain to be fully elucidated. Hopefully, this paper will inspire further research using pre-modern data to identify the central reason behind this historical pattern.

1.6 Conclusion

Agricultural feminisation—defined as women comprising more than half of the national agricultural workforce—is not a generalised trend but rather a rare exception. At the cross-country level, my research finds no evidence to support the claim that women are massively replacing men in agricultural jobs. Instead, the dominant pattern worldwide is that men are the overwhelming majority of agricultural workers (see Figure 1.2 in Appendix A, p. 42). This research thus challenges the argument that structural change leads to a feminisation of agriculture (Slavchevska et al., 2019) and contributes to the discussion arguing that the feminisation of agriculture is a myth that needs to be debunked (Kawarazuka et al., 2022). In addition, the research contributes to the literature by presenting evidence from more than 170 economies showing a clear cross-country pattern: the lower the percentage of agricultural jobs, the lower the percentage of women in the agricultural workforce. This discovery suggests an opposite hypothesis that warrants further investigation: economic development and structural transformation may have triggered a defeminisation of agriculture in some countries. Nevertheless, the extensive evidence from Mexico that is presented in this research suggests that such an interpretation is not necessarily true.

Mexico's structural transformation unfolded alongside a persistently low share of women in agricultural activities for more than a century. Therefore, there is no evidence to support the agricultural feminisation hypothesis in Mexico, but it is also not possible to argue that Mexico experienced a defeminisation of agriculture as part of its structural transformation process. The findings simply indicate that Mexican agriculture has been male-dominated since the time when the country was mostly agrarian. Hence, this study opens a new research avenue to examine whether some countries have shown low female participation in agriculture since their agrarian origins, or whether others have experienced a defeminisation of agriculture during their structural transformation process. This reframes the agricultural feminisation debate, as it seems unlikely that structural transformation in developing countries could lead to women becoming the predominant workforce in the agricultural sector.

In addition to the discussion about the feminisation of the agricultural sector, this research makes other contributions to various strands of literature. First, the study's value-added lies in providing an integrated, evidence-based evaluation of competing theories and hypotheses about the participation of women in agricultural labour markets within a single national setting. Furthermore, it bridges multiple research streams—structural transformation and technological change, local labour market structures, gendered dynamics of labour supply and labour demand,

as well as agricultural economic history—while synthesising them into a unified framework to deepen our understanding of female labour participation in agriculture. Finally, the study draws on multiple data sources to present a clear global pattern that situates Mexico within a broad cross-country picture, along with an in-depth national analysis based on labour force surveys, agricultural census data, remittance statistics from the central bank, and historical population records.

On the other hand, it is also essential to acknowledge that the results of the empirical analysis are strictly descriptive and should not be interpreted causally. It is not possible to assert that a local labour market oriented to agriculture causes low female labour force participation in Mexico. In addition, the historical factors that may explain women's persistently low participation in Mexican agriculture remain underexplored and warrant further investigation.

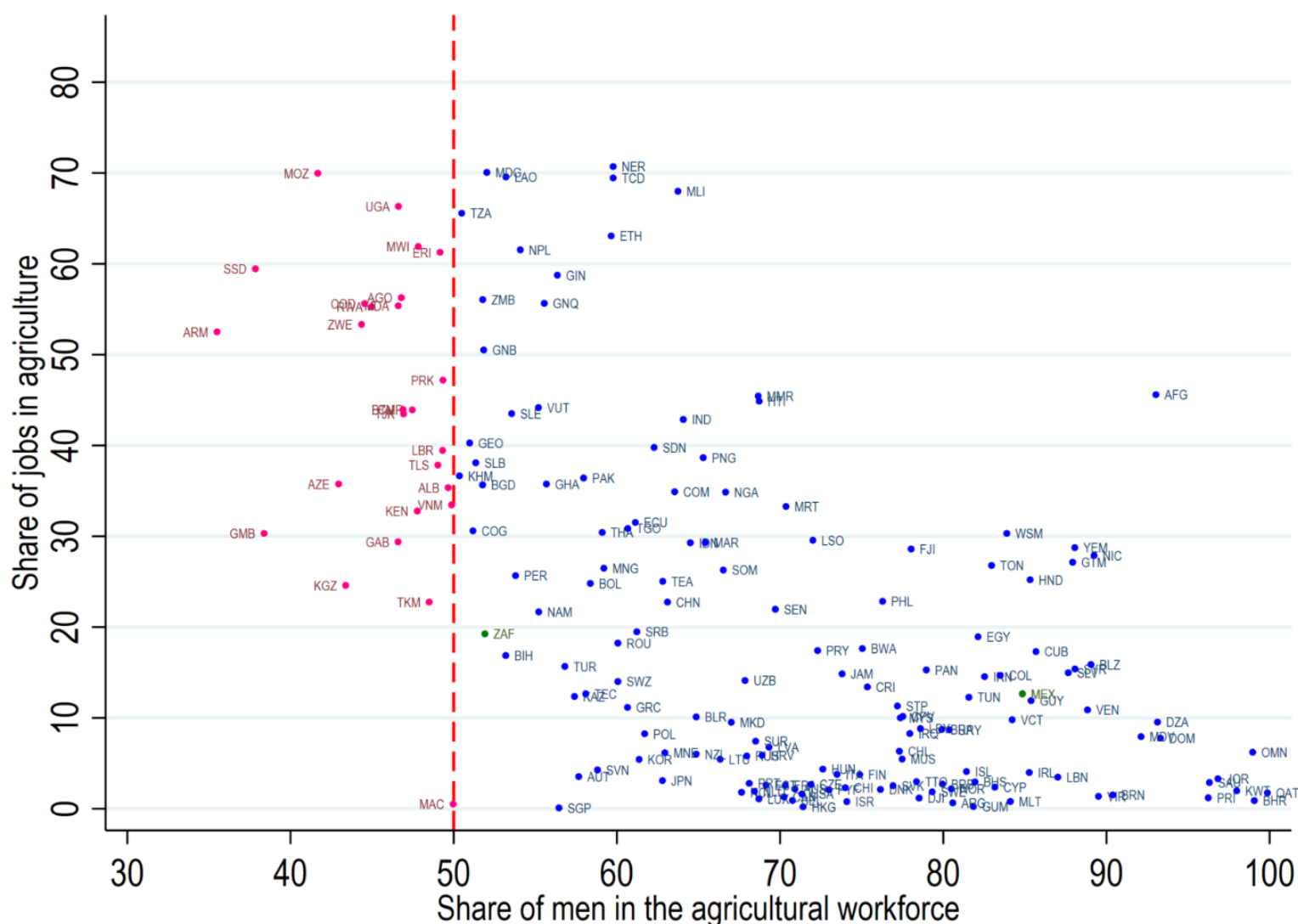
To conclude, it is important to discuss the relevance and the policy implications of this research. The analysis makes clear that policies aimed at increasing women's employment in rural Mexico should not rely on agriculture itself as the primary source of new opportunities. Expanding non-farm jobs in rural areas is the most promising pathway for increasing women's labour force participation. Rural development strategies must therefore focus on diversifying local economies beyond agriculture. Nevertheless, agricultural policy should remain attentive to ensure that women are not systematically excluded from access to farm credit or land rights.

These findings are directly relevant to the UN's Sustainable Development Goals on Gender Equality and Decent Work. Evidence from Mexico demonstrates that, in countries with historically low female agricultural participation, increasing FLPRs in rural areas requires two complementary strategies. First, identifying and addressing social norms that stigmatise women's economic activity. Second, designing policies facilitating women's employment in specific services. For instance, establishing full-time childcare centres staffed predominantly by women could professionalise rural childcare services. Such initiatives would relieve rural women of unpaid childcare burdens while generating income for centre employees, simultaneously enabling other women to redirect their time and skills towards providing additional services within their communities.

1.7 Appendix A:

Cross-country evidence

Figure 1.2 - Gender composition of the agricultural workforce and percentage of jobs in agriculture across countries (2022)



Notes:

- The figure covers more than 170 economies using World Bank Data from 2022 to capture the share of jobs in agriculture and to estimate the share of men in the agricultural workforce. Therefore, this cross-country analysis is larger than that in Doss & Gottlieb (2025), as their research considered 85 countries.
- Countries with a female majority in agriculture are highlighted in pink.⁸ Countries with a male majority are shown in blue. Mexico (MEX) and South Africa (ZAF) are highlighted in green.

⁸ The economies where women outnumber men in the agricultural workforce are: Armenia (ARM, 64.5%), South Sudan (SSD, 62.15%), The Gambia (GMB, 61.62%), Mozambique (MOZ, 58.33%), Azerbaijan (AZE, 57.05%), Kyrgyz Republic (KGZ, 56.62%), Zimbabwe (ZWE, 55.65%), Democratic Republic of Congo (COD, 55.46%), Rwanda (RWA, 55.03%), Gabon (GAB, 53.41%), Moldova (MDA, 53.4%), Uganda (UGA, 53.39%), Angola (AGO, 53.21%), Bhutan (BTN, 53.1%), Tajikistan (TJK, 53.07%), Cameroon (CMR, 52.54%), Kenya (KEN, 52.23%), Malawi (MWI, 52.17%), Turkmenistan (TKM, 51.5%), Timor-Leste (TLS, 50.97%), Eritrea (ERI, 50.84%), Liberia (LBR, 50.68%), North Korea (PRK, 50.66%), Albania (ALB, 50.34%), Viet Nam (VNM, 50.13%), and Macao SAR, China (MAC, 50.03%). Please note that, in many cases, the gender composition of the agricultural workforce is roughly 50:50. Therefore, these statistics should be interpreted with caution, as the predominance of women over men in some of these countries is not considerable.

- Doss & Gottlieb (2025), using data from 85 countries, found a few cases where women outnumber men in the agricultural workforce: Albania (ALB), Armenia (ARM), Cameroon (CMR), Kenya (KEN), Malawi (MWI), Rwanda (RWA) and Zimbabwe (ZWE).
- Slavchevska et al. (2016) identified a female predominance in agriculture in Armenia (ARM), Azerbaijan (AZE), Bhutan (BTN), The Gambia (GMB), Tajikistan (TJK), Uganda (UGA), Vietnam (VNM) and Zimbabwe (ZWE).
- Bryceson (2019) finds more women than men in agriculture in Ghana (GHA), Zambia (ZMB), Mali (MLI), Tanzania (TZA), Rwanda (RWA), Kenya (KEN), Malawi (MWI), and Uganda (UGA).
- The estimates presented in the figure above also indicate that there are more women than men in the agricultural workforce of these countries. Hence, the estimates of this study are consistent with those of the aforementioned authors.
- The data for Mexico obtained from the World Bank and presented in this figure are consistent with the information derived from ENOE surveys. INEGI estimates that in 2024, Mexico's agricultural workforce comprises 14% women and 86% men. This is consistent with my estimations using World Bank data from 2022 indicating 15% women and 85% men. Regarding the share of agricultural jobs in Mexico, the World Bank estimates around 13% in 2022, while INEGI reports approximately 11% in 2024.
- The figure suggests a clear trend: the lower the percentage of jobs in agriculture, the lower the proportion of women in the agricultural workforce. This pattern may be interpreted as a defeminisation of agriculture due to the structural transformation process, where economic development reduces agricultural employment and women's participation in the sector. Nevertheless, confirming that hypothesis requires a deeper analysis that is outside of the scope of this research. In fact, the evidence presented in this paper suggests that women's participation in agriculture has remained persistently low in Mexico since 1900. Hence, caution is advised while interpreting this figure.
- The methodology used to obtain these estimations is explained in the notes to Table 1.3 (p. 44).

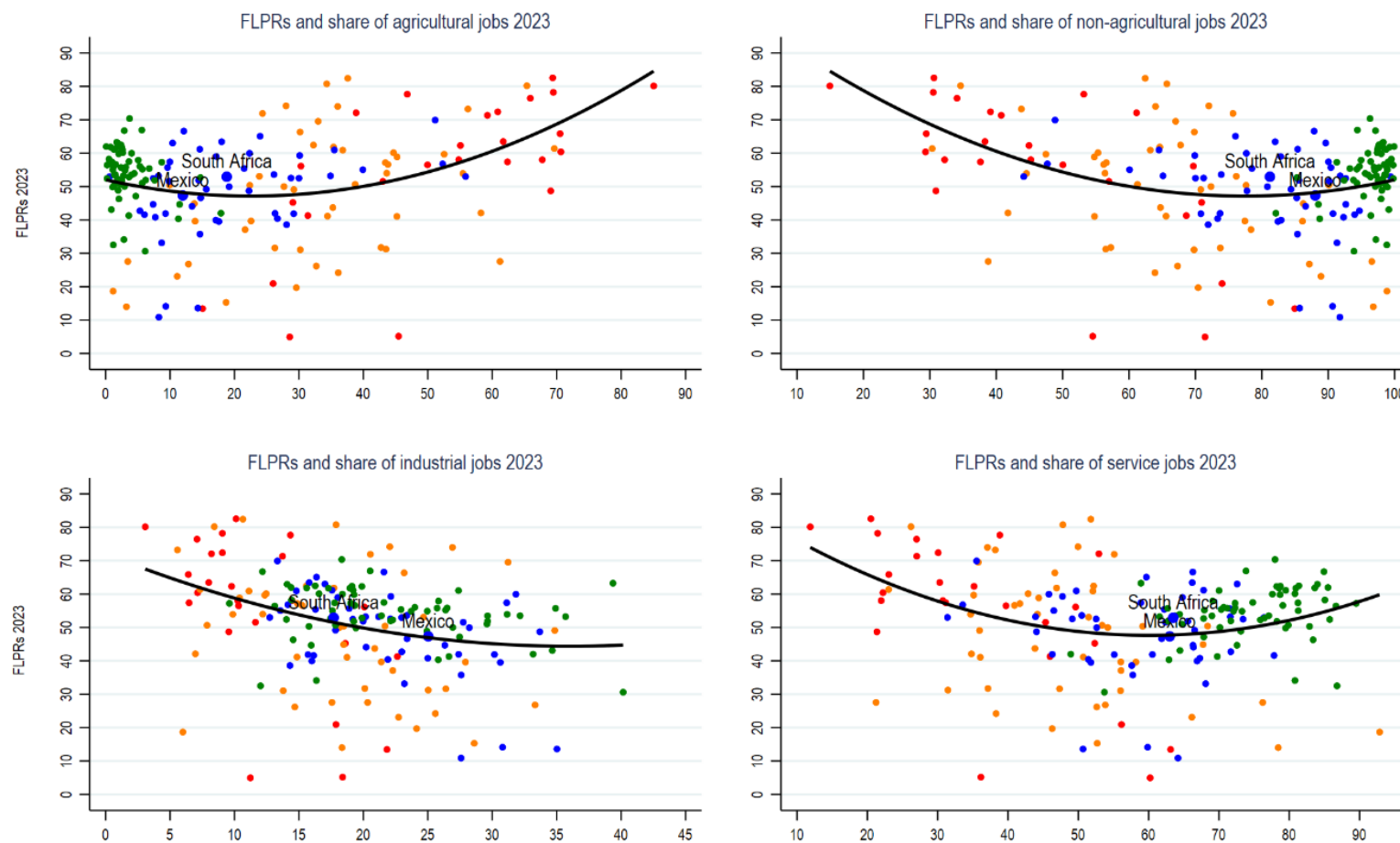
Table 1.3 - World Bank Data to estimate the sex composition of the agricultural workforce (2022)

	Sub-Saharan Africa (SSF)	South Asia (SAS)	Europe & Central Asia (ECS)	East Asia & Pacific (EAS)	North America (NAC)	Middle East & North Africa (MEA)	Latin America & Caribbean (LCN)
Labor force, total (thousands) [SL.TLF.TOTL.IN.2022]	468,906,660	733,876,869	424,090,194	1,261,873,526	189,549,748	158,934,273	316,016,483
Labor force, female (% of total labor force) [SL.TLF.TOTL.FE.ZS.2022]	46.05	26.32	45.62	44.24	46.40	19.57	41.73
Labor force, male (% of total labor force) [Estimated by the author]	53.95	73.68	54.38	55.76	53.60	80.43	58.27
Female labor force, total (thousands) [Estimated by the author]	215,934,349	193,184,012	193,478,420	558,308,805	87,955,043	31,101,174	131,867,716
Male labor force, total (thousands) [Estimated by the author]	252,972,311	540,692,857	230,611,774	703,564,721	101,594,705	127,833,099	184,148,767
Employment in agriculture, female (% of female employment) [SL.AGR.EMPL.FE.ZS.2022]	51.56	59.18	6.59	19.49	0.98	12.83	7.62
Employment in agriculture, male (% of male employment) [SL.AGR.EMPL.MA.ZS.2022]	51.83	35.36	8.45	25.75	2.10	13.26	17.83
Female agricultural labor force, total (thousands) [Estimated by the author]	111,340,627	114,322,026	12,752,572	108,823,073	863,443	3,989,304	10,054,380
Male agricultural labor force, total (thousands) [Estimated by the author]	131,127,963	191,185,574	19,488,876	181,201,738	2,138,527	16,950,861	32,833,619
Agricultural labor force, total (thousands) [Estimated by the author]	242,468,590	305,507,600	32,241,448	290,024,811	3,001,969	20,940,165	42,887,999
Employment in agriculture, female (% of total agricultural labor force) [Estimated by the author]	45.92	37.42	39.55	37.52	28.76	19.05	23.44
Employment in agriculture, male (% of total agricultural labor force) [Estimated by the author]	54.08	62.58	60.45	62.48	71.24	80.95	76.56

Notes to Table 1.3:

- *The information presented in the table is based on World Bank Data for 2022.*
- *Indicators drawn from World Bank databases are reported with their official code in brackets, while self-calculations are indicated explicitly as [Estimated by the author].*
- *The total labour force (in thousands) is taken directly from the World Bank indicator [SL.TLF.TOTL.IN.2022].*
- *The proportion of the labour force that is female comes from the World Bank indicator [SL.TLF.TOTL.FE.ZS.2022].*
- *The proportion of the labour force that is male is not directly reported by the World Bank but is inferred by subtracting the female share from 100, based on the proportion of women in the total labour force. This is shown in brackets as [Estimated by the author].*
- *Based on these data, the absolute number of female and male workers in the labour force is calculated and reported in brackets as [Estimated by the author].*
- *Women employed in agriculture as a percentage of the total female workforce is taken from [SL.AGR.EMPL.FE.ZS.2022]*
- *Men employed in agriculture as a percentage of the total male workforce is taken from [SL.AGR.EMPL.MA.ZS.2022].*
- *Based on these data, it is possible to estimate the total number of male and female workers in agriculture.*
- *Summing female and male agricultural workers provides the total agricultural labour force, also marked as [Estimated by the author].*
- *The percentage of men in agriculture is calculated by multiplying the total agricultural labour force by 100 and dividing it by the number of men employed in agriculture.*
- *The percentage of women in agriculture is calculated by multiplying the total agricultural labour force by 100 and dividing it by the number of women employed in agriculture.*
- *For analytical clarity, the results report only the seven regional categories defined by the World Bank.*
- *Presenting these calculations for more than 170 countries would yield a table that is difficult to interpret; therefore, the results are presented at the regional level to illustrate the steps followed to estimate the percentage of men and women in the agricultural workforce of more than 170 countries.*

Figure 1.3 - Female labour participation rates and sectoral shares of employment across countries (2023)



Notes:

Income group classification: Red = low income; Orange = lower-middle income; Blue = upper-middle income; Green = High income.
Data obtained from the World Bank Development Indicators.

1.8 Appendix B:

Within-country evidence

Table 1.4 – Results of probit regressions: likelihood that Mexican men are participating in the workforce depending on the sectoral distribution of employment at the municipal level.

	(1) Non-Agricultural	(2) Agriculture	(3) Industry	(4) Services	(5) Industry & services	(6) Non-Agricultural (Squared)
Share of non-agricultural jobs at the municipal level	-0.00340*** (0.00062)					-0.00760*** (0.00210)
Share of agricultural jobs at the municipal level		0.00340*** (0.00062)				
Share of industrial jobs at the municipal level			-0.00118* (0.00069)		-0.00278*** (0.00076)	
Share of service jobs at the municipal level				-0.00240*** (0.00067)	-0.00360*** (0.00073)	
Share of non-agricultural jobs at the municipal level (squared)						0.00003** (0.00002)
Control variables - Individual	Yes	Yes	Yes	Yes	Yes	Yes
Control variables - Household	Yes	Yes	Yes	Yes	Yes	Yes
Control variables - Municipal	Yes	Yes	Yes	Yes	Yes	Yes
Year/quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Clustered standard errors	Yes	Yes	Yes	Yes	Yes	Yes
Constant	-1.45655*** (0.26413)	-1.79702*** (0.24817)	-1.99222*** (0.24755)	-1.67675*** (0.26847)	-1.44535*** (0.27219)	-1.37175*** (0.26728)
Observations	486,484	486,484	486,484	486,484	486,484	486,484

Notes

Base category for state fixed effects: Mexico City

Base category for time fixed effects: First quarter of 2019.

Clustered standard errors at the municipal level.

Standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 1.5 – Predicted probabilities that women (and men) are part of the labour force depending on the share of non-agricultural jobs in their municipality.

%	Women		Men	
	Share of non-agricultural jobs at the municipal level	Share of non-agricultural jobs at the municipal level	Share of non-agricultural jobs at the municipal level	Share of non-agricultural jobs at the municipal level
		(<i>squared</i>)		(<i>squared</i>)
0	0.4255*** (0.0185)	0.3175*** (0.0210)	0.9855*** (0.0021)	0.9892*** (0.0022)
10	0.4478*** (0.0166)	0.3827*** (0.0175)	0.9842*** (0.0020)	0.9866*** (0.0021)
20	0.4703*** (0.0146)	0.4429*** (0.0143)	0.9828*** (0.0019)	0.9838*** (0.0020)
30	0.4929*** (0.0126)	0.4961*** (0.0117)	0.9814*** (0.0018)	0.9809*** (0.0019)
40	0.5155*** (0.0106)	0.5409*** (0.0099)	0.9798*** (0.0017)	0.9781*** (0.0018)
50	0.5380*** (0.0086)	0.5769*** (0.0086)	0.9781*** (0.0015)	0.9755*** (0.0017)
60	0.5605*** (0.0066)	0.6042*** (0.0075)	0.9763*** (0.0014)	0.9732*** (0.0017)
70	0.5827*** (0.0048)	0.6229*** (0.0061)	0.9743*** (0.0013)	0.9713*** (0.0015)
80	0.6047*** (0.0034)	0.6334*** (0.0045)	0.9723*** (0.0012)	0.9700*** (0.0014)
90	0.6263*** (0.0030)	0.6358*** (0.0032)	0.9701*** (0.0012)	0.9693*** (0.0012)
100	0.6476*** (0.0038)	0.6303*** (0.0043)	0.9677*** (0.0014)	0.9693*** (0.0014)
Obs.	538,433	538,433	486,484	486,484

Notes:

The average marginal probabilities were obtained after considering the means of the other covariates estimated from the regression analysis.

Standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

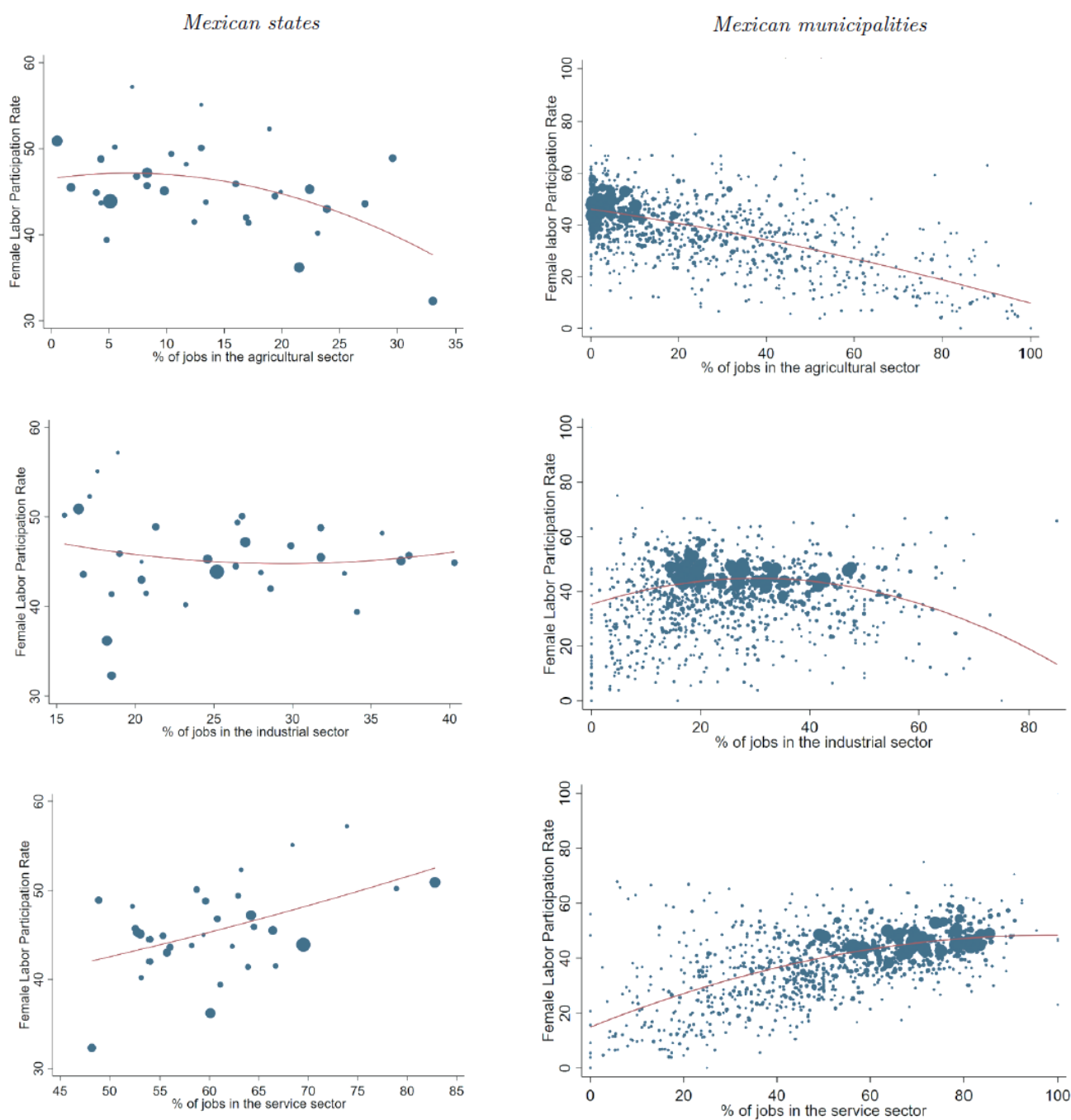
Table 1.6 – Selection ratios

WOMEN			
	Coefficient (without controls)	Coefficient (with controls)	δ Altonji Selection Ratio
Share of non-agricultural jobs	0.0086231*** (0.00030)	0.00553*** (0.00053)	1.79
Share of agricultural jobs	-0.0086231*** (0.00030)	-0.00553*** (0.00053)	1.79
Share of industrial jobs	.0029139*** (0.00069)	0.00223*** (0.00058)	3.26
Share of service jobs	.008776*** (0.00036)	0.00388*** (0.00056)	0.79

Notes:

Altonji et al. (2005) show that selection ratios offer an informative diagnostic for assessing the relative importance of omitted variable bias in empirical estimates. These ratios quantify how much stronger selection on unobserved determinants would need to be, relative to selection on observed controls, to overturn the estimated results. Therefore, they use this metric to assess the sensitivity of their results to unobserved confounding. For instance, they interpret a ratio of 3.55 as indicating that attributing the estimated result entirely to selection bias seems highly unlikely, whereas a ratio of 0.08 suggests that the result could be easily explained by unobserved selection. In general terms, they argue that once the econometric model incorporates a powerful set of control variables, the regression results with selection ratios greater than one are unlikely to be driven by omitted variable bias, since the unobserved variables would have to be as strong as the observables to rule out the results.

Figure 1.4 – FLPRs and share of jobs in each economic sector.
(Mexican states and Mexican municipalities, first quarter of 2019)



Notes:

*The size of the dots at the state level reflects the total population in each state.
The size of the dots at the municipal level reflects the total number of surveys carried out in each municipality.*

Figure 1.5 – Gender composition of the labour force in agriculture, industry, and services in Mexico (2005–2019)

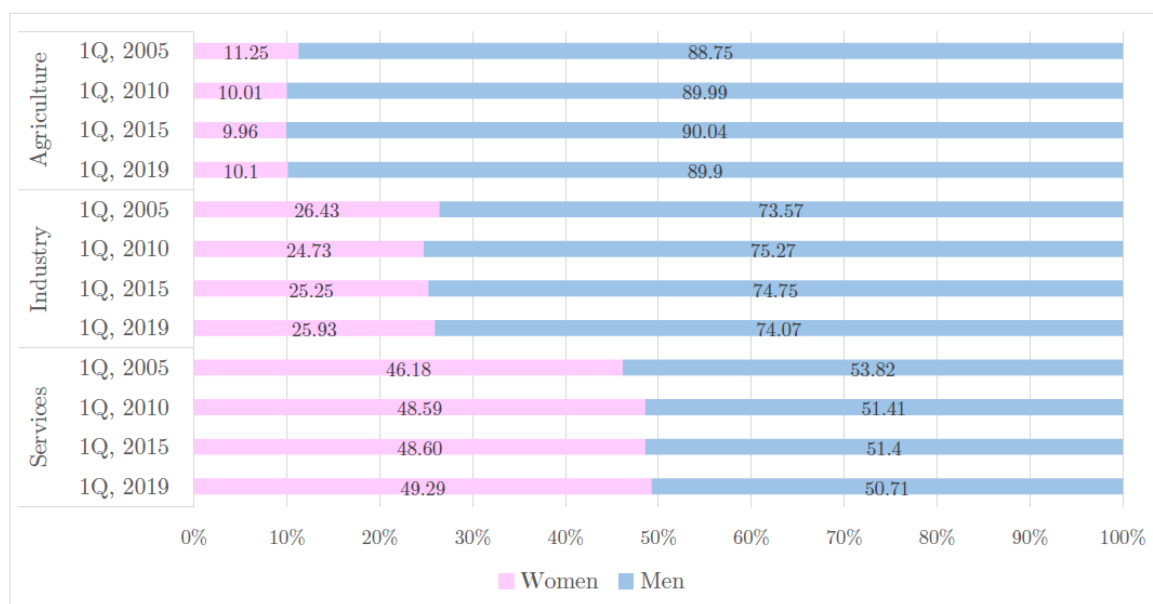


Figure 1.6 – Distribution of the male and female labour force by sector of employment in Mexico (2005–2019)

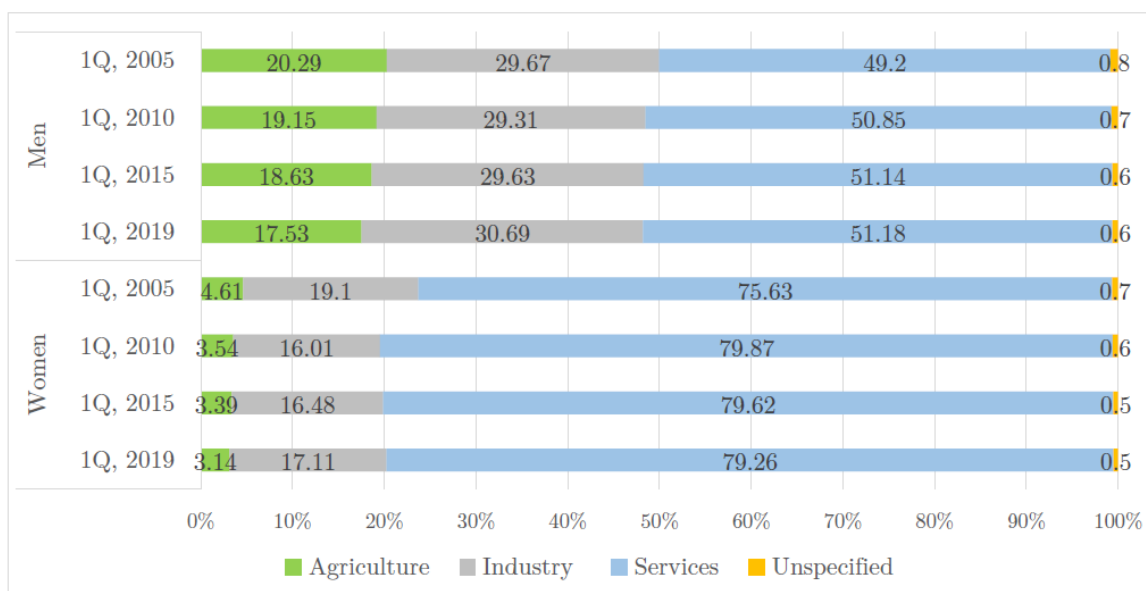
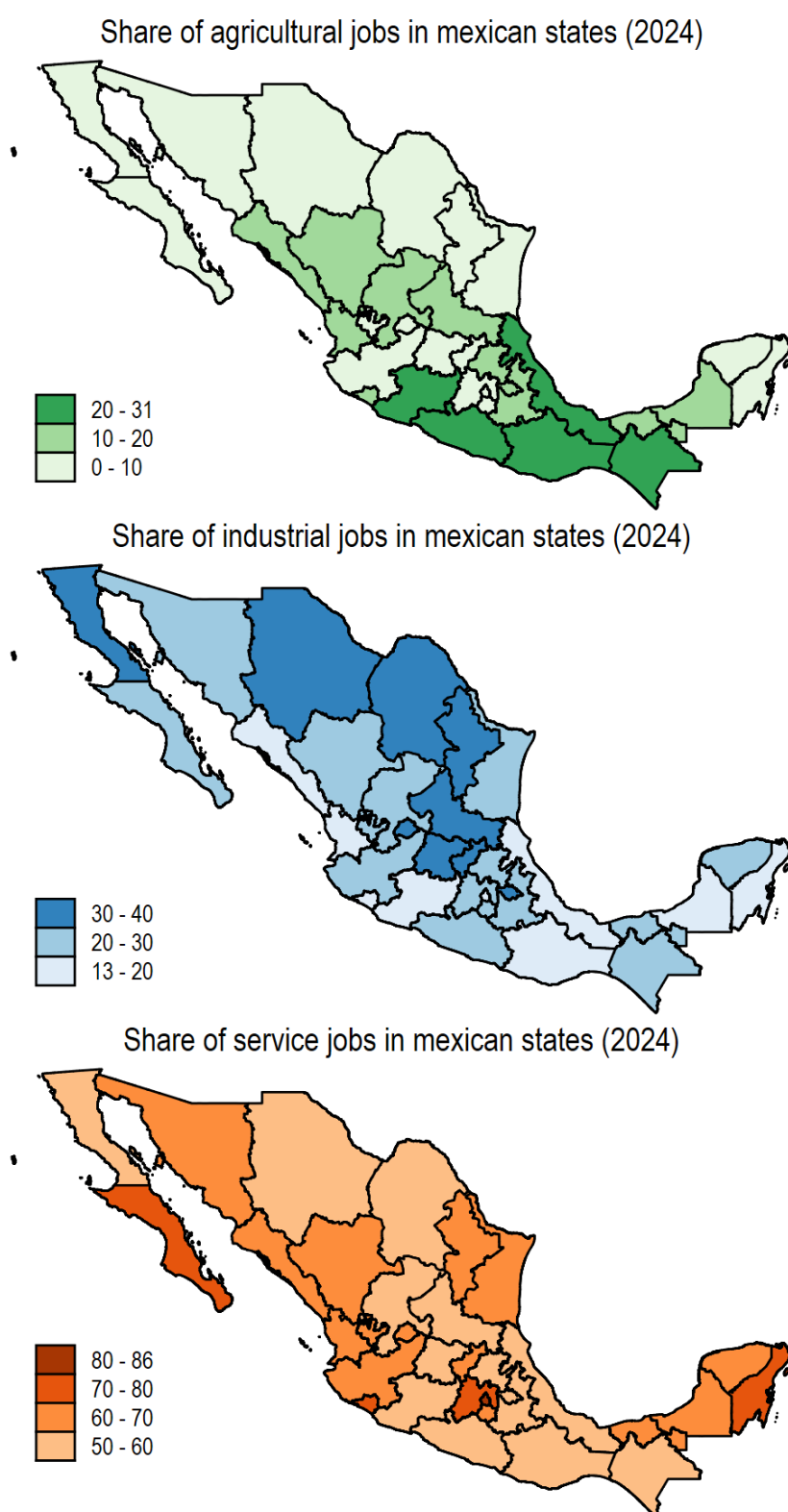


Figure 1.7 – Maps of sectoral distribution of employment in Mexican states (2024)



Note:

Sectoral shares were obtained from the ENOE survey of the first quarter of 2024. Details on the sectoral distribution of employment in Mexican states are reported in Table 1.7.

Table 1.7 - Sectoral distribution of employment in Mexican states (2024)

State	Share of jobs in agriculture	Share of jobs in industry	Share of jobs in services
Aguascalientes	4.2	31.7	63.7
Baja California	3.7	31.8	59.6
Baja California Sur	4.4	22.3	73.2
Campeche	17.7	19	62.9
Coahuila	2.3	39.9	57.3
Colima	10.9	16.4	71.8
Chiapas	28	21.7	50.2
Chihuahua	6.5	37.8	53.1
Ciudad de México	0.7	13.6	85.2
Durango	11.7	26.5	61.3
Guanajuato	9.9	32.4	57.5
Guerrero	23.9	21.1	55
Hidalgo	16.2	23.6	60
Jalisco	9.3	25.8	64.6
México	3.5	24.1	72.1
Michoacán	22.2	20	57.7
Morelos	8.1	23	68.7
Nayarit	16.1	16.2	67.3
Nuevo León	1.7	34.5	63.7
Oaxaca	30.2	18.9	50.7
Puebla	18.1	28.3	53.4
Querétaro	3.7	32.6	63.6
Quintana Roo	4.9	16.6	78.2
San Luis Potosí	11.2	32.7	55.8
Sinaloa	14.1	20	65.3
Sonora	9.1	27.4	61.6
Tabasco	14.5	21.6	63.7
Tamaulipas	6.1	29	63.2
Tlaxcala	10.5	34.1	55.1
Veracruz	21.3	18.5	59.9
Yucatán	8.3	26.9	64.7
Zacatecas	18	22	59.5

Note: These estimations were obtained using data from the ENOE household survey.

Figure 1.8 – Average size of agricultural production units in Mexican states, in hectares (2022)



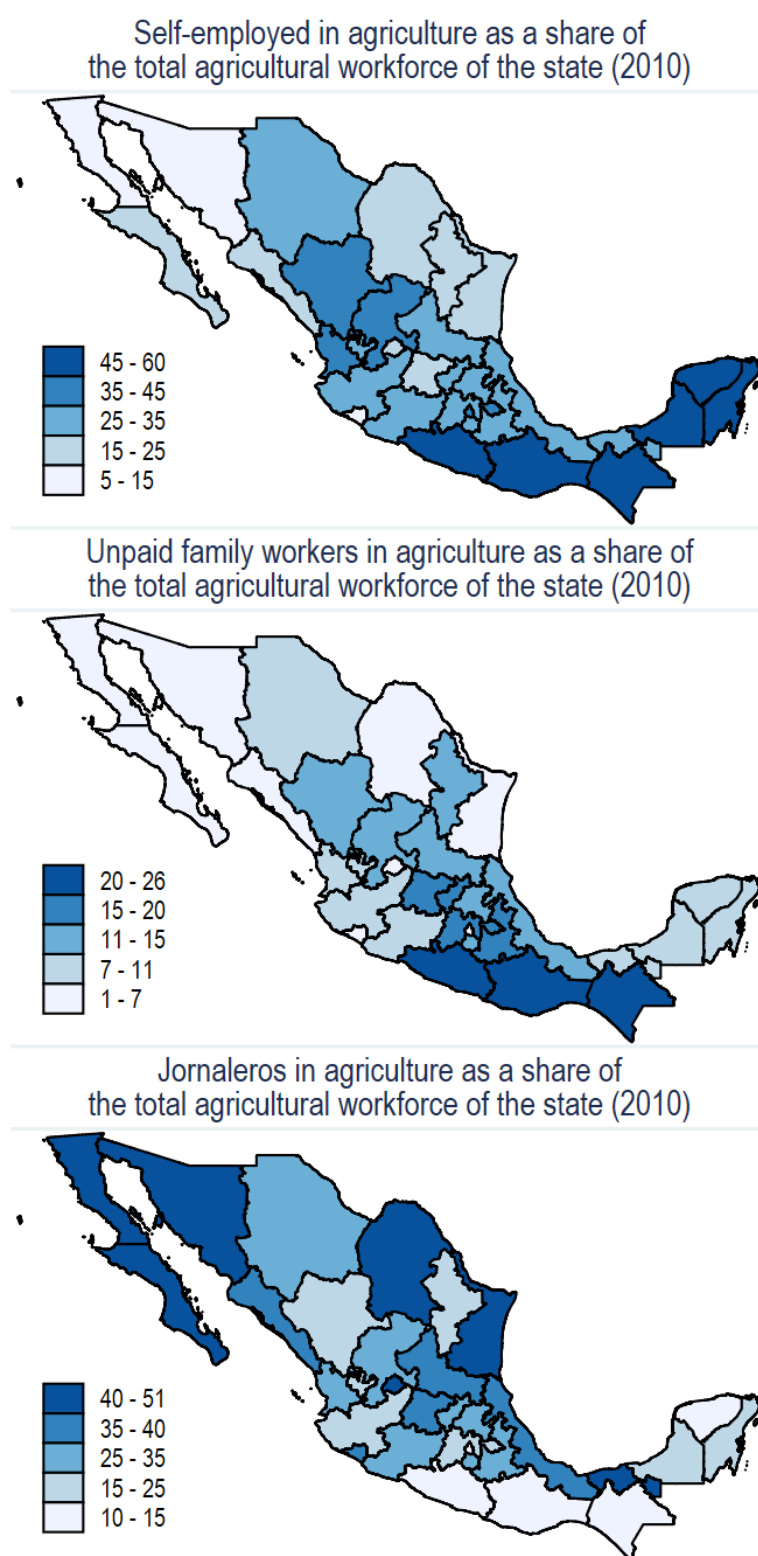
Source: Mexico's Agricultural Census (2022)

Table 1.8 – Average size of agricultural production units in Mexican states, in hectares (2022)

State	Agricultural Production Units	Area in hectares	Average size
Aguascalientes	19,648	234,352	11.93
Baja California	7,777	918,530	118.11
Baja California Sur	4,606	635,605	138.00
Campeche	53,018	1,537,470	29.00
Coahuila de Zaragoza	26,189	4,486,611	171.32
Colima	14,476	276,650	19.11
Chiapas	492,046	3,206,899	6.52
Chihuahua	85,628	11,016,862	128.66
Ciudad de México	11,410	18,071	1.58
Durango	91,425	2,031,362	22.22
Guanajuato	159,838	1,338,565	8.37
Guerrero	299,587	2,048,013	6.84
Hidalgo	214,374	769,061	3.59
Jalisco	179,732	3,469,928	19.31
México	399,260	860,807	2.16
Michoacán de Ocampo	228,603	2,114,894	9.25
Morelos	63,768	202,452	3.17
Nayarit	74,702	830,676	11.12
Nuevo León	28,097	2,388,009	84.99
Oaxaca	380,061	1,813,213	4.77
Puebla	457,612	1,359,268	2.97
Querétaro	42,471	270,893	6.38
Quintana Roo	27,295	458,012	16.78
San Luis Potosí	138,109	1,632,115	11.82
Sinaloa	95,201	1,849,779	19.43
Sonora	38,039	8,987,244	236.26
Tabasco	125,011	1,569,347	12.55
Tamaulipas	65,328	3,519,754	53.88
Tlaxcala	88,466	249,693	2.82
Veracruz de Ignacio de la Llave	521,503	4,446,485	8.53
Yucatán	58,036	1,095,850	18.88
Zacatecas	137,818	2,566,555	18.62
Mexico (Total)	4,629,134	68,203,023	14.73

Source: Mexico's Agricultural Census (2022)

Figure 1.9 – Maps of the employment composition of the agricultural workforce across Mexican states (2010)



Note: Jornaleros is a term referring to those people working in agriculture who are paid on a daily basis and are hired under informal arrangements, as they do not have access to medical insurance or a pension scheme. Additional details about the statistics presented in this figure can be found in Table 1.9.

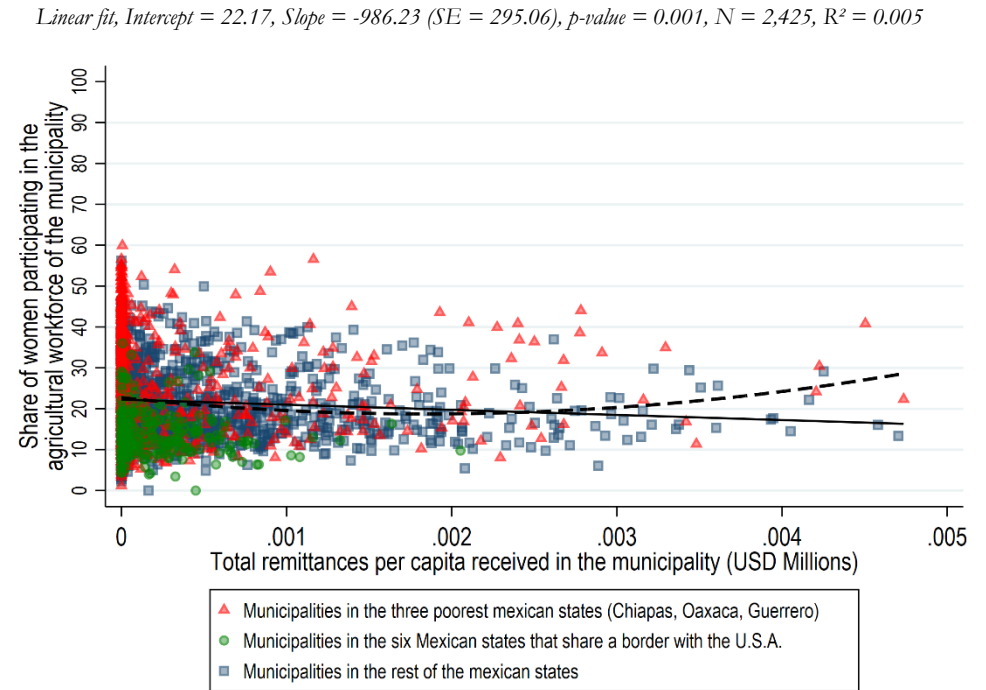
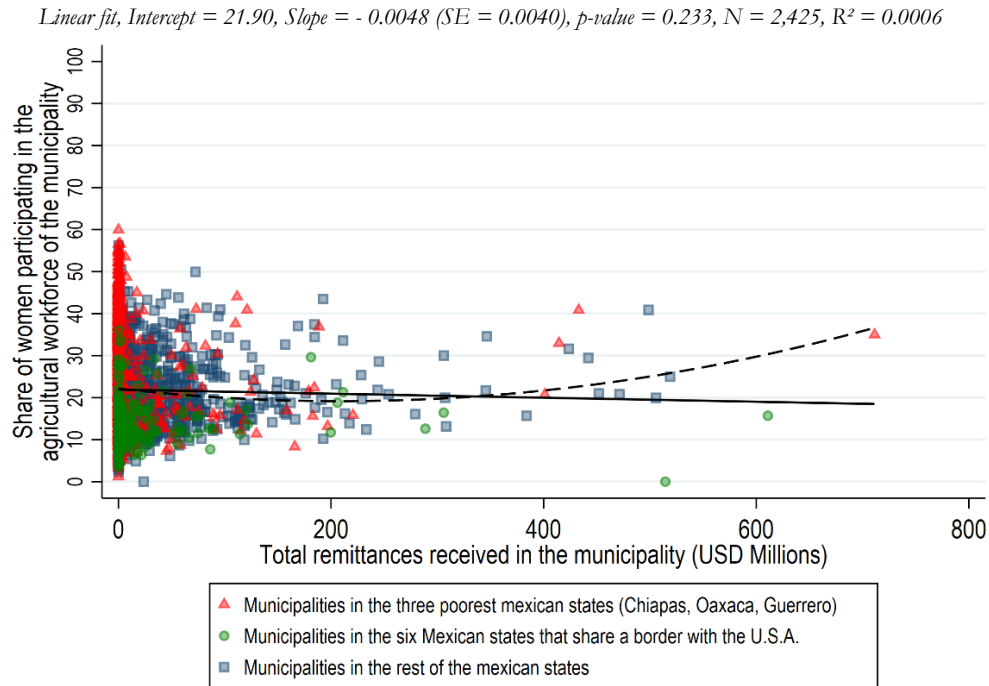
Table 1.9 - Employment composition of the agricultural workforce across Mexican states (2010)

State	Employers	Jornaleros	Self-employed	Unpaid family workers	Employees	Helpers	Unspecified	Total
Aguascalientes	3.77	42.73	19.56	6.31	20.88	5.59	1.17	100.00
Baja California	2.60	50.75	5.41	1.19	37.77	1.54	0.75	100.00
Baja California Sur	5.19	40.22	21.59	3.26	26.04	2.50	1.21	100.00
Campeche	1.53	21.03	47.32	10.19	14.70	4.30	0.93	100.00
Coahuila	2.09	42.92	20.14	6.41	22.60	4.59	1.24	100.00
Colima	4.38	37.35	14.74	2.66	33.18	6.24	1.46	100.00
Chiapas	0.72	14.59	56.60	20.16	4.36	1.62	1.95	100.00
Chihuahua	3.88	28.75	33.11	8.86	20.41	3.50	1.48	100.00
Ciudad de México	6.91	11.80	38.38	5.36	28.64	6.57	2.35	100.00
Durango	2.93	20.50	36.69	12.76	20.67	4.60	1.86	100.00
Guanajuato	1.94	37.22	24.05	15.39	16.03	4.15	1.22	100.00
Guerrero	0.58	14.62	51.20	25.55	2.86	2.60	2.59	100.00
Hidalgo	1.01	33.63	34.71	14.95	8.62	6.02	1.06	100.00
Jalisco	3.36	20.06	29.57	7.90	31.20	6.71	1.20	100.00
México	1.71	23.28	33.85	17.84	14.61	7.28	1.45	100.00
Michoacán	2.66	34.55	30.62	8.30	17.48	4.83	1.54	100.00
Morelos	2.84	30.98	32.20	11.57	16.29	3.87	2.24	100.00
Nayarit	3.26	32.83	37.83	7.96	13.78	3.34	0.99	100.00
Nuevo León	2.74	23.90	24.86	11.63	29.03	5.57	2.28	100.00
Oaxaca	0.76	13.34	51.64	24.62	4.82	2.85	1.98	100.00
Puebla	1.19	31.59	34.68	16.73	8.22	5.80	1.79	100.00
Querétaro	2.07	29.04	30.37	15.23	17.16	4.51	1.63	100.00
Quintana Roo	3.79	16.68	53.24	10.64	11.59	2.35	1.71	100.00
San Luis Potosí	0.86	36.84	33.06	13.88	10.16	4.08	1.12	100.00
Sinaloa	2.73	38.63	24.84	4.01	25.52	2.33	1.94	100.00
Sonora	3.56	48.39	14.12	1.92	28.43	2.57	1.01	100.00
Tabasco	2.22	44.05	29.55	8.15	11.04	3.05	1.93	100.00
Tamaulipas	4.45	42.65	23.02	5.27	19.63	3.41	1.57	100.00
Tlaxcala	0.55	20.48	43.83	18.73	7.27	8.07	1.07	100.00
Veracruz	2.24	36.78	34.81	11.08	9.46	4.12	1.50	100.00
Yucatán	1.09	11.90	54.02	9.21	17.89	4.48	1.40	100.00
Zacatecas	2.20	29.63	36.08	13.20	11.81	5.82	1.26	100.00

Notes:

*These estimations were obtained using data from the Mexican Population Census of 2010.
The statistics present the distribution of the total agricultural workforce in each Mexican state, broken down into different types of agricultural labour.*

Figure 1.10 – Remittances and share of women in agriculture (2022)



Notes:

The first scatterplot shows the total remittances received in each Mexican municipality (in USD millions).

The second scatterplot shows the remittances per capita at the municipal level (in USD millions).

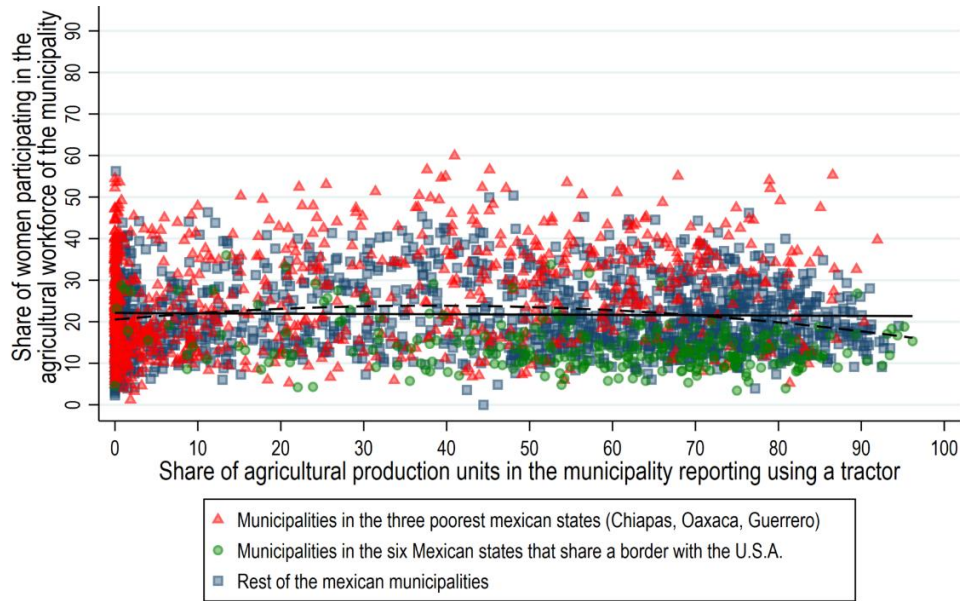
Data on remittances were sourced from the official statistics of Mexico's Central Bank.

Per capita remittances are estimated by dividing total remittances by the municipal population.

Share of women in the municipal agricultural workforce is sourced from Mexico's Agricultural Census (2022).

Figure 1.11 – Use of tractors and share of women in agriculture (2022)

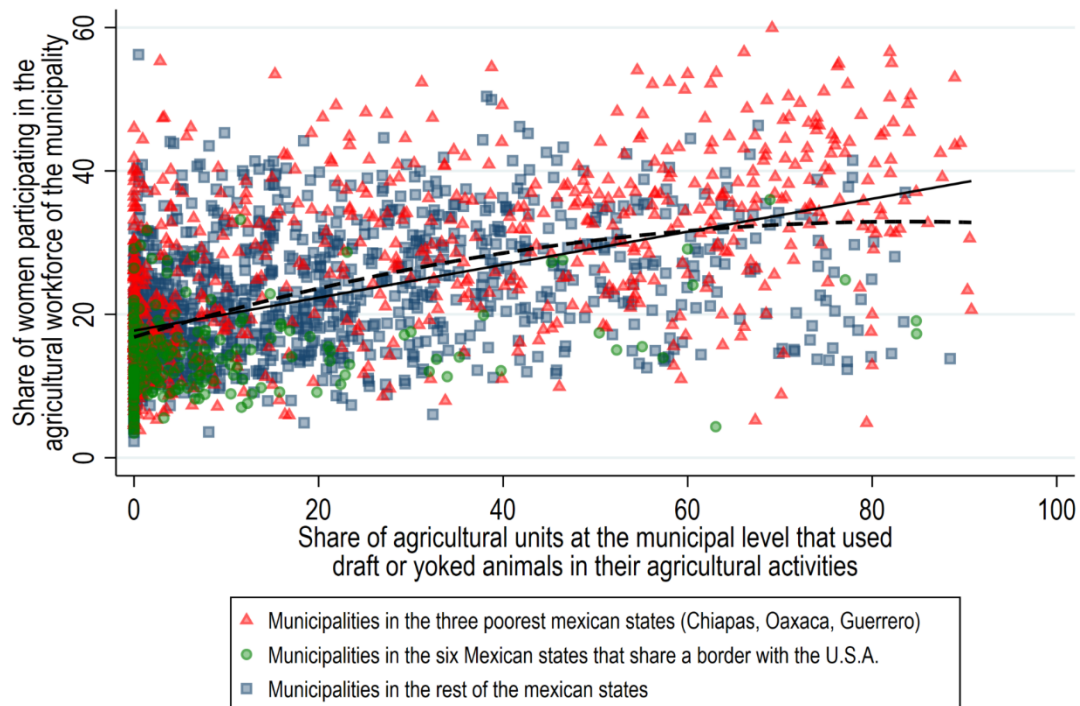
Linear fit, Intercept = 22.11, Slope = -0.0078 (SE = 0.0070), p-value = 0.266, N = 2,456, R² = 0.0005



Note: Data obtained from Mexico's Agricultural Census (2022).

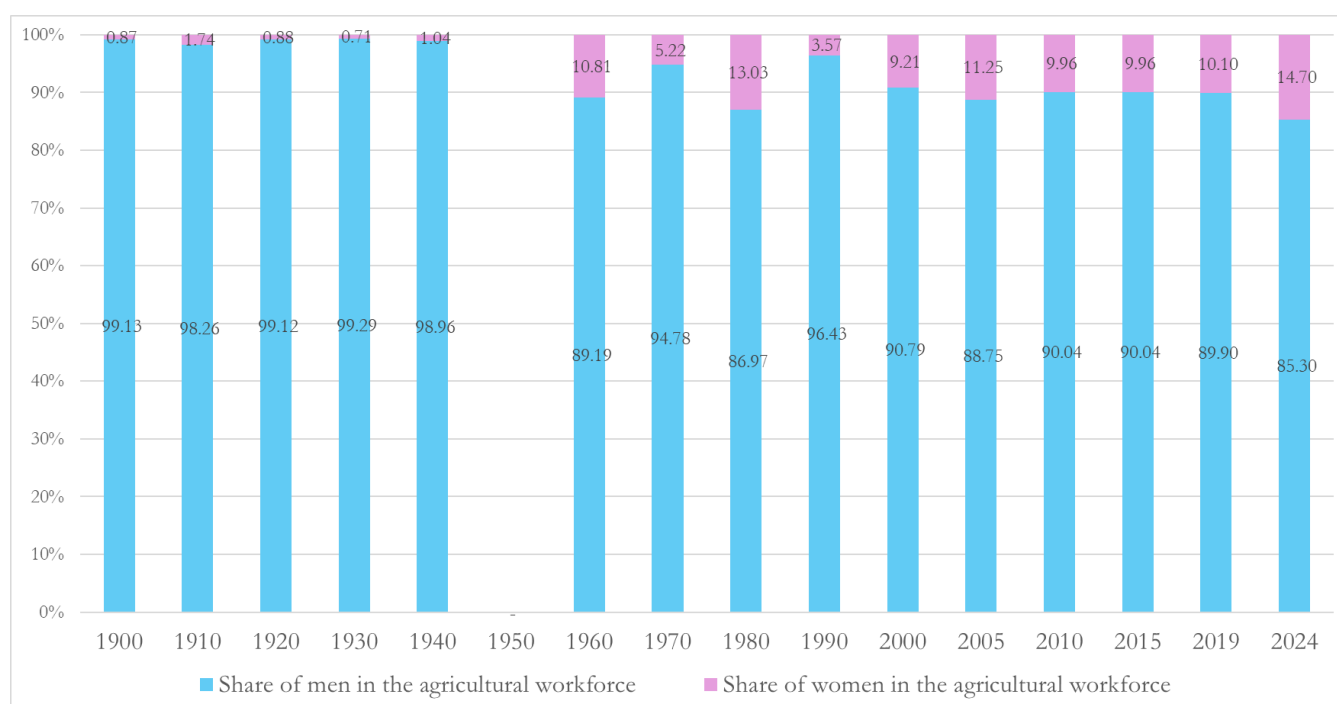
Figure 1.12 – Use of draught or yoked animals and share of women in agriculture (2022)

Linear fit, Intercept = 17.72, Slope = 0.23 (SE = 0.0079), p-value = 0.000, N = 2,456, R² = 0.275



Note: Data obtained from Mexico's Agricultural Census (2022).

Figure 1.13 - Gender composition of the agricultural labour force in Mexico (1900–2024)



Notes:

Data disaggregated by sex is not available for 1950.

Additional statistics related to this Figure are presented in Table 1.10

Table 1.10 - Gender composition of the agricultural workforce in Mexico (1900–2024)

Year	Total agricultural workforce	Men in agriculture	Women in agriculture	Share of men	Share of women
1900	3,176,571	3,148,944	27,627	99.13	0.87
1910	3,596,157	3,533,557	62,600	98.26	1.74
1920	3,488,102	3,457,571	30,531	99.12	0.88
1930	3,626,278	3,600,645	25,633	99.29	0.71
1940	3,830,871	3,791,007	39,864	98.96	1.04
1950	.	.	.	.	.
1960	6,143,540	5,479,642	663,898	89.19	10.81
1970	5,103,519	4,836,865	266,654	94.78	5.22
1980	5,700,860	4,958,146	742,714	86.97	13.03
1990	5,300,114	5,110,964	189,150	96.43	3.57
2000	5,338,299	4,846,900	491,399	90.79	9.21
2005	6,082,242	5,397,845	684,397	88.75	11.25
2010	5,910,159	5,321,731	588,428	90.04	9.96
2015	6,308,756	5,680,218	628,538	90.04	9.96
2019	6,436,643	5,786,285	650,358	89.90	10.10
2024	6,367,745	5,431,626	936,119	85.30	14.70

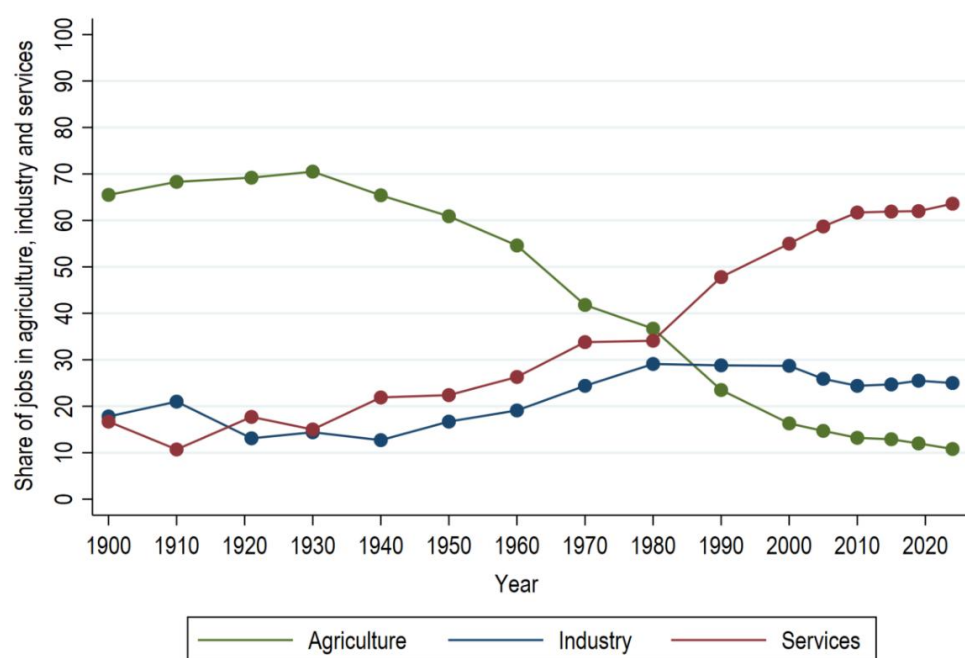
Notes:

Data from 1900 to 2000 were obtained from the corresponding population censuses of that year.

The census of 1950 does not report information about the gender composition of the agricultural workforce.

Data from 2005 to 2024 were drawn from the first-quarter releases of the ENOE survey.

Figure 1.14 - Structural transformation of Mexico's labour markets (1900–2024)



Note: Additional statistics related to this Figure are presented in Table 1.11.

Table 1.11 - Structural transformation of Mexico's labour markets (1900–2024)

Year	Total workforce	Share in agriculture	Share in industry	Share in services
1900	4,807,385	65.5	17.8	16.7
1910	5,265,453	68.3	21.0	10.7
1921	5,044,432	69.2	13.1	17.7
1930	5,141,331	70.5	14.4	15.0
1940	5,858,116	65.4	12.7	21.9
1950	7,917,127	60.9	16.7	22.4
1960	11,250,225	54.6	19.1	26.3
1970	12,207,532	41.8	24.4	33.8
1980	15,515,934	36.7	29.1	34.1
1990	22,599,541	23.5	28.8	47.8
2000	32,717,631	16.3	28.7	55.0
2005	43,099,847	14.7	25.9	58.7
2010	47,223,973	13.2	24.4	61.7
2015	51,176,166	12.9	24.7	61.9
2019	55,578,352	12.0	25.5	62.0
2024	60,663,120	10.8	25.0	63.6

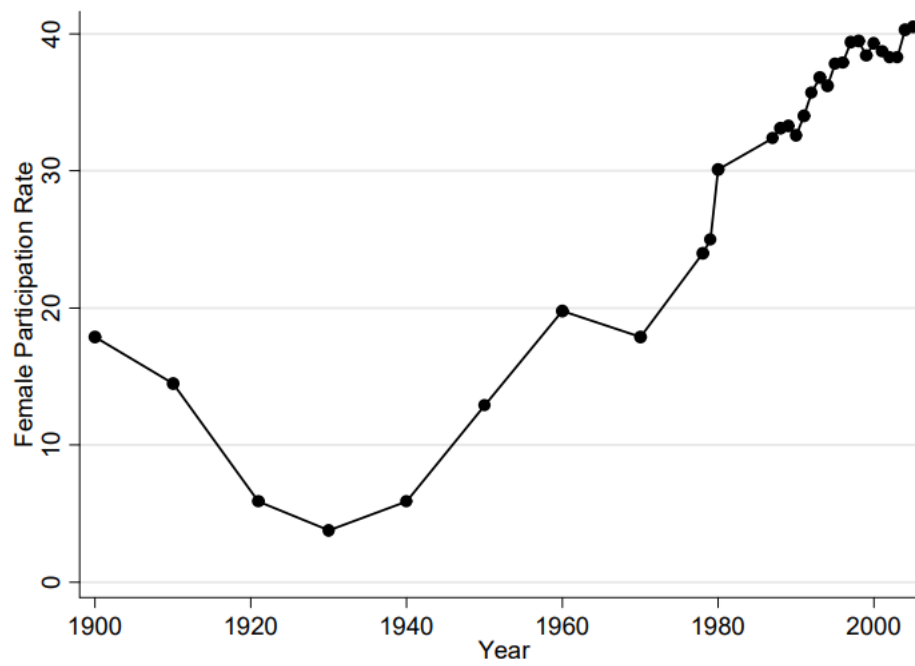
Notes:

Data from 1900 to 2000 were obtained from the corresponding population censuses covering those years.

Data from 2005 to 2024 were drawn from the first-quarter releases of the ENOE survey.

Sectoral shares derived from the ENOE survey may not sum to exactly 100% thanks to the inclusion of an additional category corresponding to a small fraction of unspecified economic activities.

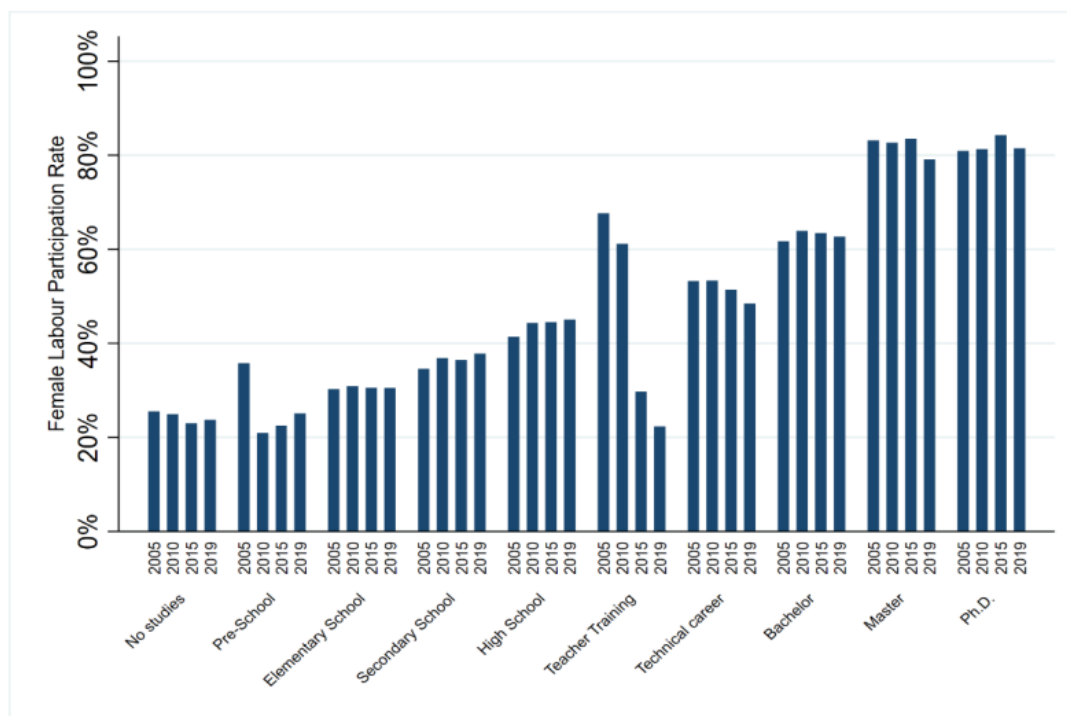
Figure 1.15 – Historical female labour participation rate in Mexico (1900–2005)



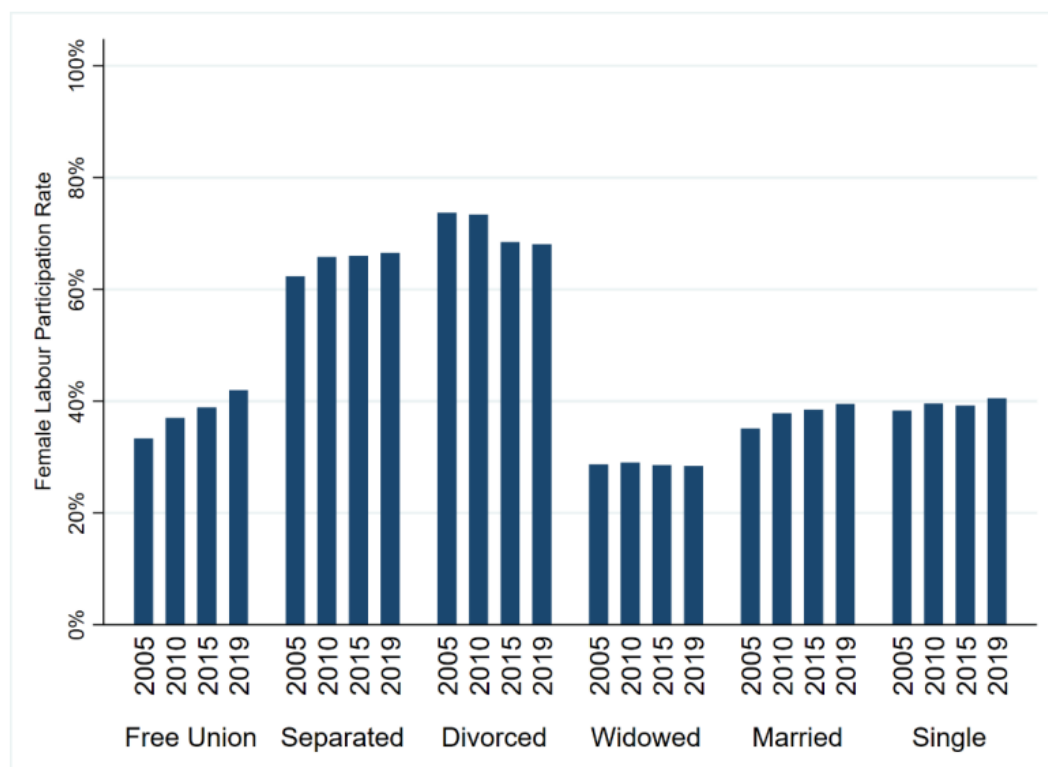
Note: This figure was obtained from Arceo-Gomez & Campos-Vazquez (2010)

Figure 1.16 – Control variables

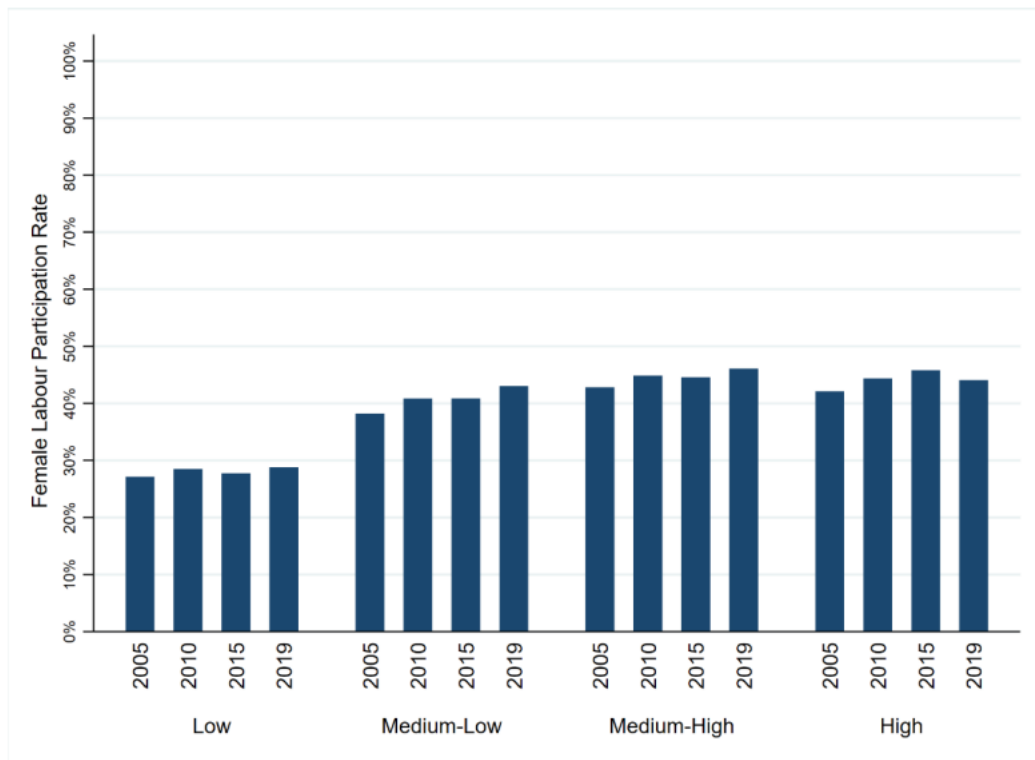
a) FLPRs based on different education levels (2005–2019)



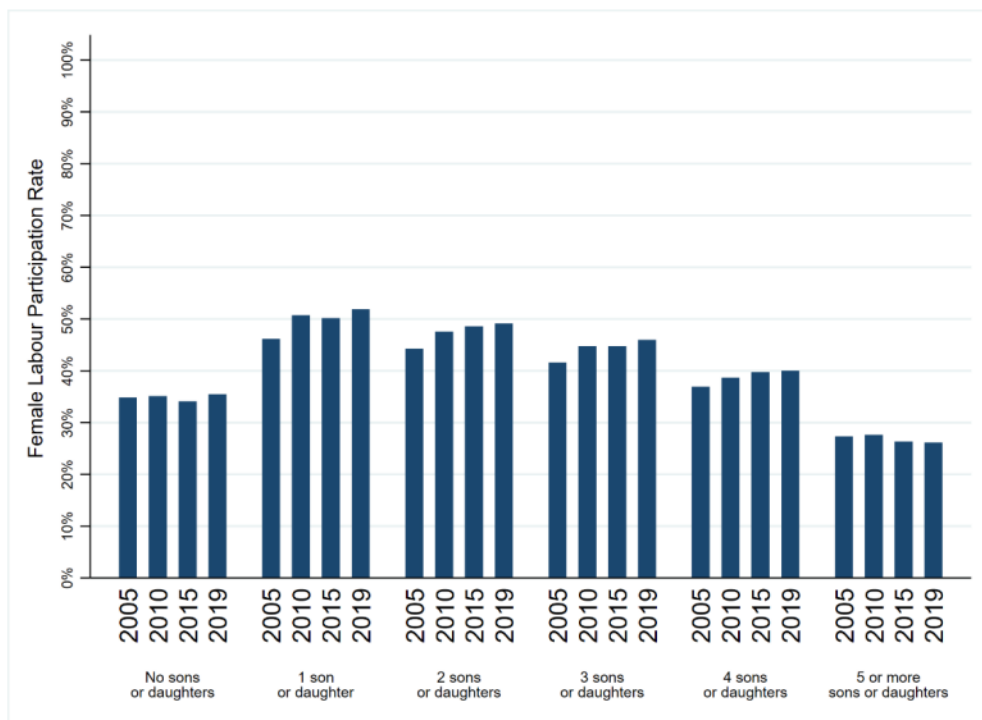
b) FLPRs based on marital status (2005–2019)



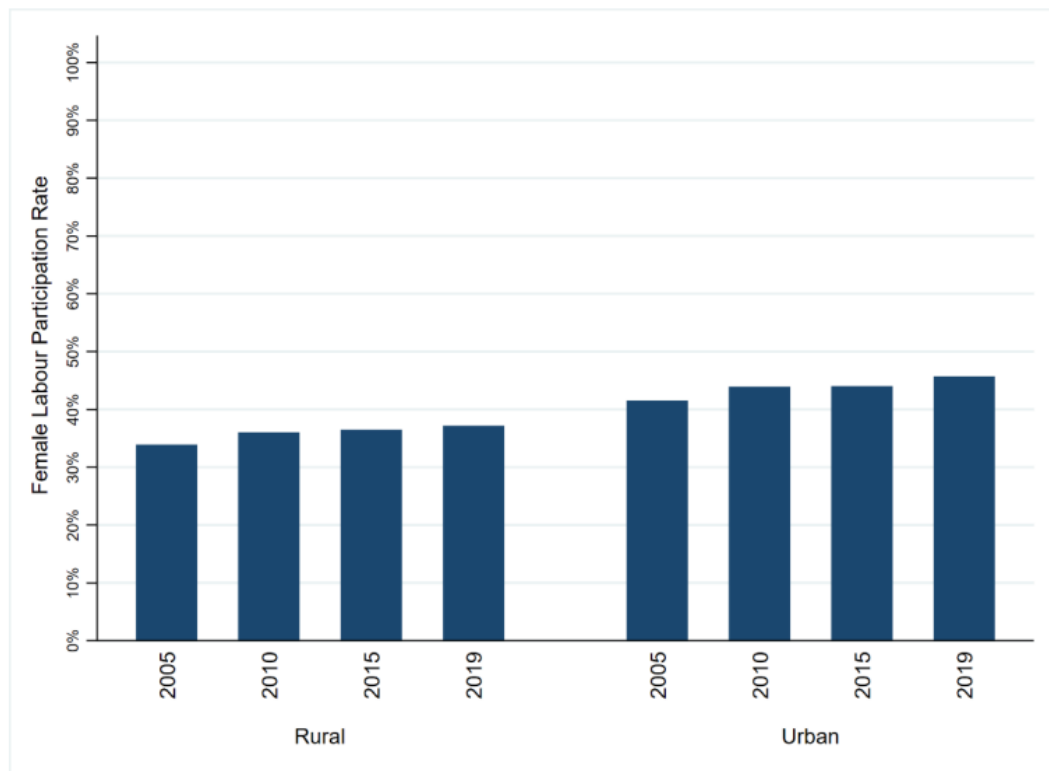
c) FLPRs based on socioeconomic stratum (2005–2019)



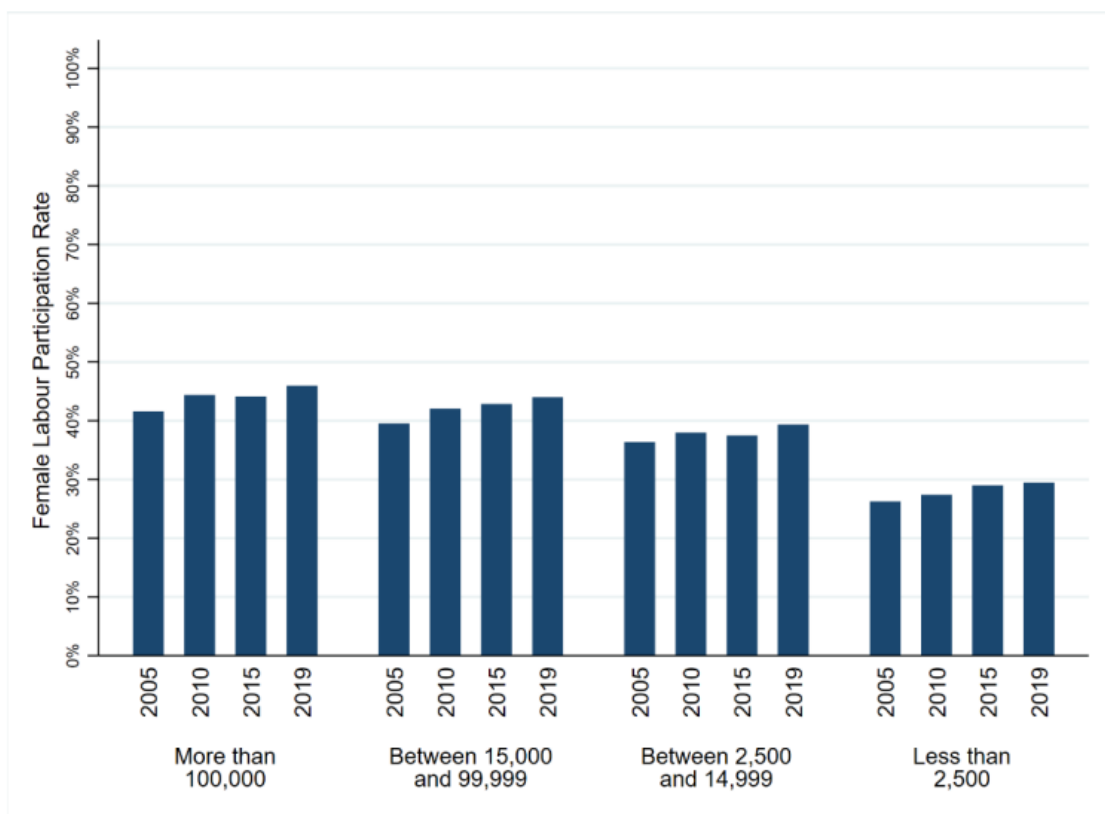
d) FLPRs based on number of children (2005–2019)



e) FLPRs in rural and urban areas (2005–2019)



f) FLPRs based on the population size in the household location (2005–2019)



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